Weatherford

Four Barriers Corrosion Evaluation

Company:	SAUDI ARAMCO
Well:	HRDH_1702_1
Field:	HARADH
Country:	SAUDI ARABIA
Log Date:	15-May-2026
Report Delivery Date:	22-May-2026

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DISCLAIMER: The end of tubing depth could not be confirmed from the recorded data due to the tool becoming stuck at 6990 ft (DF). Following PE approval, MSP was applied at the end of tubing. Multiple up/down passes were performed in the open-hole section at different logging speeds in an attempt to regain tool movement. Due to the over-pull event encountered at the end of tubing, data acquisition between 6975 ft and 6925 ft was not recorded.

Log correlation was performed using the client-provided Gamma Ray (GR) log over the interval from 6900 ft to 6500 ft

1. Objectives

- Evaluate the current completion condition of **HRDH_1702_1** and assess the need for any workover maintenance.
- Locate, identify, and quantify metal loss within the **18 5/8”, 10 3/4” 7” casing strings, 8 5/8”, 7 5/8" liner strings and 4 1/2” tubing string** from the specified bottom log interval at **7233ft** up to the surface.
- Identify wellbore temperature anomalies.

2. Conclusions

- Minor metal loss detected across the **18 5/8” casing string.**
- Minor metal loss detected across the **10 3/4” casing string.**
- Minor metal loss detected across the **8 5/8” liner string.**
- Moderate metal loss detected across the **7 5/8” liner string.**
- Intensive metal loss detected across the **7” casing string.**
- Minor metal loss detected across the **4 1/2" tubing string.**
- Temperature anomaly observed across the logging interval.
- Excellent data quality & repeatability.

Pipe OD	Max WL	Grade of Metal Loss	Hot Spot Depth (ft)
18 5/8" CSG	5.1%	B	18.3
10 3/4" CSG	6.2%	B	457.1
8 5/8" LNR	8.7%	B	1345.6
7 5/8" LNR	11.2%	C	4027.0
7" CSG	31.2%	D	4180.4
4 1/2" TBG	7.0%	B	4180.2

Table 1 Hot Spots

3. Summary and Other Remarks

- Well Name: **HRDH_1702_1** ARBD OIL OBSERVATION.
- Original Completion: {{OG}}.
- Last workover: {{WKO}}.
- Log Date: {{logdate}}
- LOG CORRELATED TO GR LOG PROVIDED BY THE CLIENT AT THE INTERVAL 6900 - 6500 ft.
- Tool stuck at 6990 ft (DF) and MSP was applied at the end of tubing after PE approval. We went up and down multiple times on OH sections with different speeds (60, 80, 100, 140, 160 ft/min).
- Regarding the over-pull event at the end of tubing, we missed recording the data from 6975 to 6925 ft with total agreement from PE and Aramco supervision, and we could not go to this depth again.
- All procedures performed were in total agreement with the client.
- Rigless operation, logged with GOWell ePDT tool, processed with SmartLog.
- Metal loss is expressed as percentage (%) and represents average circumferential metal loss for the tubing and casing strings, surrounding the logging tool.

4. One Page Summary

HRDH_1702_1

Well Information

ePDT Pipe Metal loss Summary

RIGLESS
ARBD OIL OBSERVATION.
•Log Date: 15-May-2026.
•Original Completion Date:10-Mar-2004.
•Latest Workover Date: 07-Mar-2012 #1.
•LOG CORRELATED TO GR LOG PROVIDED BY THE CLIENT AT

Completion Strings

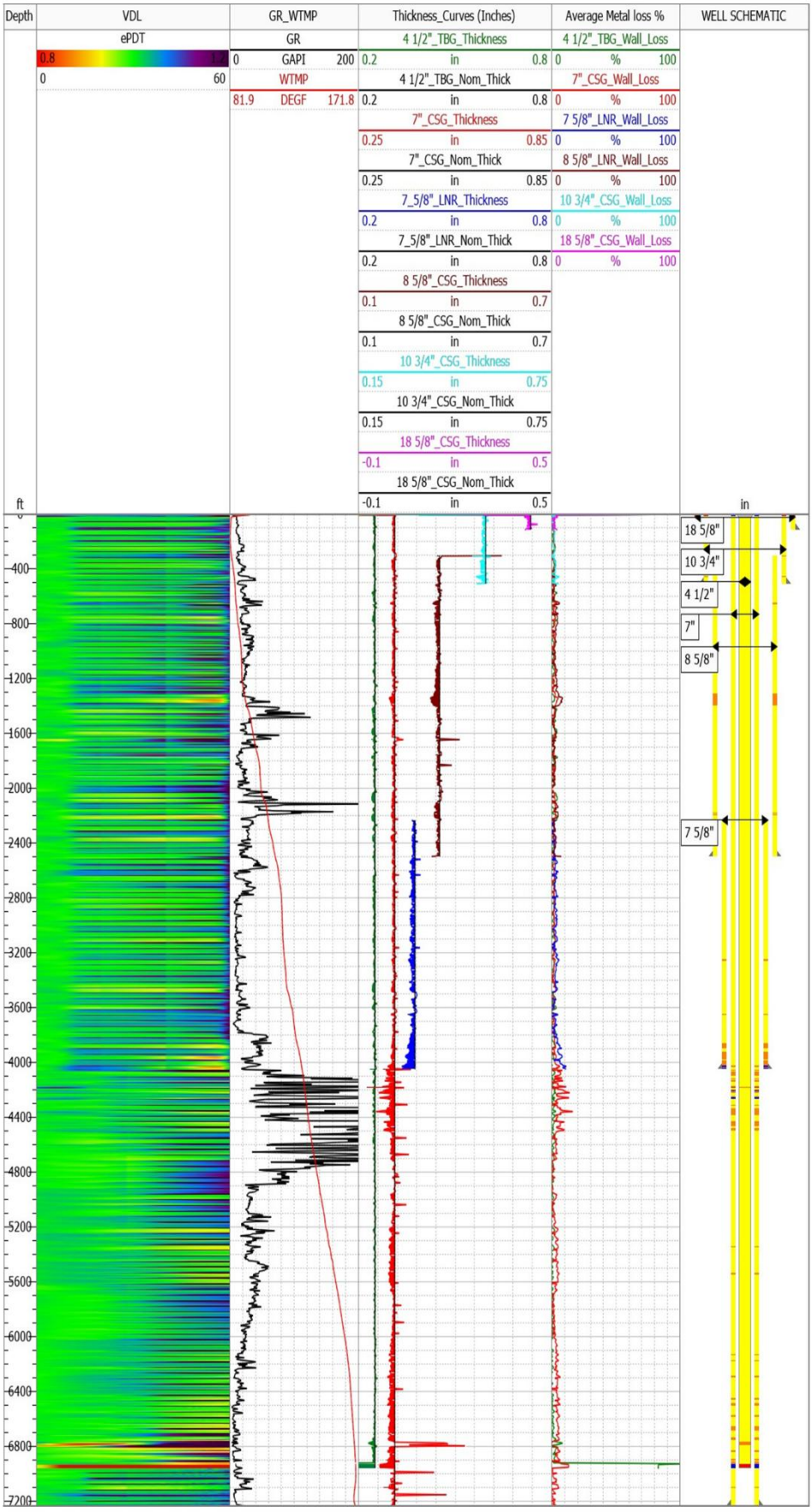
Pipe OD (inch)	Weight (ppf)	Pipe Interval (ft)		Thick_Nom (inch)
18 5/8" CSG	87.5	5	117	0.435
10 3/4" CSG	60.7	5	510	0.545
8 5/8" LNR	32	307	2499	0.352
7 5/8" LNR	29.7	2234.5	4050	0.375
7" CSG	26	5	7233	0.362
4 1/2" TBG	11.6	5	Didn't Detect	0.250

Hot Spot Summary

Pipe OD (inch)	Max WL	Grade of Metal loss	Hotter Spot Depth (ft)
18 5/8" CSG	5.1%	B	18.3
10 3/4" CSG	6.2%	B	457.1
8 5/8" LNR	8.7%	B	1345.6
7 5/8" LNR	11.2%	C	4027.0
7" CSG	31.2%	D	4180.4
4 1/2" TBG	7.0%	B	4180.2

Conclusions

•Minor metal loss detected across the 18 5/8" casing string.
•Minor metal loss detected across the 10 3/4" casing string.
•Minor metal loss detected across the 8 5/8" liner string.
•Moderate metal loss detected across the 7 5/8" liner string.
•Intensive metal loss detected across the 7" casing string.
•Minor metal loss detected across the 4 1/2" tubing string.
•Temperature anomaly observed across the logging interval.



5. Saudi Aramco Metal Loss Classification

The table below classifies metal loss categories to evaluate integrity of tubulars, as per Saudi Aramco well integrity standards.

Grade	Maximum Wall Loss	Level
A	0-5%	Within Tool Accuracy Calculation
B	5%-10%	Minor
C	10%-20%	Moderate
D	>=20%	Intensive
E	<0%	Special Higher Weight Joint

Table 2 Metal Loss Classification

6. ePDT processing workflow

- In the table below we highlight Channels used to calculate for each pipe across one interval.
- Values of Contrast applied for each computed thickness curve aligned with the processing guidelines (K1=0.45, K2=0.55, K3=0.7 and K4 =0.9).
- Baselines shift (offset) may apply to calibrate across nominal pipe(s) intervals and values will be reported per section.
- Identify intervals with different baselines and apply zones whenever needed (additional snapshots may be added to elaborate on the anomalies seen).

Well Name	{{Well_name}}		
RIG Name	RIGLESS		
Intervals	Interval 1	Interval 2	Interval 3
Start Depth(ft)	5	117	307
End Depth(ft)	117	307	510
Tubular size & weight	4 1/2"TBG-0.250"@11.6ppf	4 1/2"TBG-0.250"@11.6ppf	4 1/2"TBG-0.250"@11.6ppf
	7"CSG-0.362"@26ppf	7"CSG-0.362"@26ppf	7"CSG-0.362"@26ppf
	10 3/4"CSG-0.545"@60.7ppf	10 3/4"CSG-0.545"@60.7ppf	8 5/8"LNR-0.352"@32ppf
	18 5/8"CSG-0.435"@87.5ppf	/	10 3/4"CSG-0.545"@60.7ppf
Interpretation Channels	18-35-50-56	18-35-50	18-35-50-56
Pipe channel response	10-26-47-55	10-26-47	10-26-47-55
Offset	0.001/-0.002/0/0.003	0.001/0/-0.002	-0.001/-0.002/0.004/0.003
Intervals	Interval 4	Interval 5	Interval 6
Start Depth(ft)	510	2234.5	2499
End Depth(ft)	2234.5	2499	4050
Tubular size & weight	4 1/2"TBG-0.250"@11.6ppf	4 1/2"TBG-0.250"@11.6ppf	4 1/2"TBG-0.250"@11.6ppf
	7"CSG-0.362"@26ppf	7"CSG-0.362"@26ppf	7"CSG-0.362"@26ppf
	8 5/8"LNR-0.352"@32ppf	7 5/8"LNR-0.375"@29.7ppf	7 5/8"LNR-0.375"@29.7ppf
	/	8 5/8"LNR-0.352"@32ppf	/
Interpretation Channels	18-35-50	18-35-50-56	18-35-53
Pipe channel response	10-26-47	10-26-47-55	10-26-47
Offset	0.001/0/-0.002	0.001/-0.002/0/0.003	0.001/0/-0.002
Intervals	Interval 7	Interval 8	
Start Depth(ft)	4050	Not Detected	
End Depth(ft)	Not Detected	7233	
Tubular size & weight	4 1/2"TBG-0.250"@11.6ppf	7"CSG-0.362"@26ppf	
	7"CSG-0.362"@26ppf	/	
Interpretation Channels	26-49	25	
Pipe channel response	18-35	18	
Offset	0.006/0.005	0.014	

Table 3 Tubulars Specification and Interpretation Channels Selection

7. Tabulated Joints with Highest Metal Loss

Table 4 Highest Metal Loss in 4 1/2" Tubing

#	Top Body (ft)	Bottom Body (ft)	Body Length (ft)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (ft)	Max Loss (%)	Grade	Damage Profile (% wall loss)
52	4954.0	4990.1	36.1	0.362	0.329	4963.3	9.2	B	
73	5803.9	5842.0	38.2	0.362	0.331	5836.3	8.6	B	
48	4809.4	4848.4	39.0	0.362	0.332	4848.3	8.3	B	
68	5599.3	5638.3	39.0	0.362	0.333	5632.3	8.0	B	

Table 5 Highest Metal Loss in 7" Casing

#	Top Body (ft)	Bottom Body (ft)	Body Length (ft)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (ft)	Max Loss (%)	Grade	Damage Profile (% wall loss)
104	3819.5	3855.0	35.5	0.395	0.253	3824.9	35.9	D	
101	3707.7	3743.9	36.1	0.395	0.255	3740.4	35.4	D	
105	3857.0	3892.3	35.3	0.395	0.266	3876.8	32.8	D	
97	3561.3	3594.8	33.5	0.395	0.292	3579.1	25.9	D	
111	4077.9	4113.2	35.3	0.395	0.293	4098.3	25.7	D	
103	3782.5	3817.5	35.0	0.395	0.297	3810.2	24.8	D	
106	3894.3	3928.7	34.4	0.395	0.299	3904.4	24.4	D	
112	4115.2	4150.2	35.0	0.395	0.303	4117.6	23.3	D	
99	3634.4	3669.3	35.0	0.395	0.303	3662.2	23.2	D	
102	3745.9	3780.5	34.7	0.395	0.308	3779.6	22.0	D	
109	4004.0	4039.0	35.0	0.395	0.317	4014.7	19.8	C	
100	3671.3	3705.7	34.4	0.395	0.320	3700.9	18.9	C	
108	3968.2	4002.0	33.8	0.395	0.335	4000.9	15.3	C	
107	3930.7	3966.2	35.5	0.395	0.336	3946.5	15.0	C	
113	4152.2	4187.4	35.3	0.395	0.344	4183.5	12.8	C	
96	3523.5	3559.3	35.8	0.395	0.346	3534.9	12.3	C	

Table 6 Highest Metal Loss in 7 5/8" Liner

#	Top Body (ft)	Bottom Body (ft)	Body Length (ft)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (ft)	Max Loss (%)	Grade	Damage Profile (% wall loss)
53	2075.8	2114.8	39.0	0.430	0.388	2076.9	9.7	B	
52	2035.9	2073.8	37.9	0.430	0.391	2070.7	9.2	B	
45	1759.1	1797.9	38.7	0.430	0.393	1786.2	8.6	B	
46	1799.9	1838.1	38.2	0.430	0.393	1804.3	8.5	B	

Table 7 Highest Metal Loss in 8 5/8" Liner

#	Top Body (ft)	Bottom Body (ft)	Body Length (ft)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (ft)	Max Loss (%)	Grade	Damage Profile (% wall loss)
8	306.2	346.7	40.5	0.435	0.393	319.2	9.6	B	
9	348.7	388.0	39.3	0.435	0.395	382.4	9.2	B	
12	474.7	514.1	39.3	0.435	0.398	510.2	8.4	B	
13	516.1	556.6	40.5	0.435	0.401	553.6	7.9	B	

Table 8 Highest Metal Loss in 10 3/4" Casing

#	Top Body (ft)	Bottom Body (ft)	Body Length (ft)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (ft)	Max Loss (%)	Grade	Damage Profile (% wall loss)
2	72.2	114.0	41.8	0.688	0.635	74.6	7.7	B	

Table 9 Highest Metal Loss in 18 5/8" Casing

#	Top Body (ft)	Bottom Body (ft)	Body Length (ft)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (ft)	Max Loss (%)	Grade	Damage Profile (% wall loss)
145	5346.1	5385.3	39.3	0.250	0.230	5346.3	7.8	B	
144	5305.1	5344.1	39.0	0.250	0.233	5308.6	7.0	B	
31	1162.8	1201.0	38.2	0.250	0.234	1189.1	6.2	B	
131	5093.7	5133.2	39.5	0.250	0.236	5100.6	5.8	B	

8. Raw Data Log Display

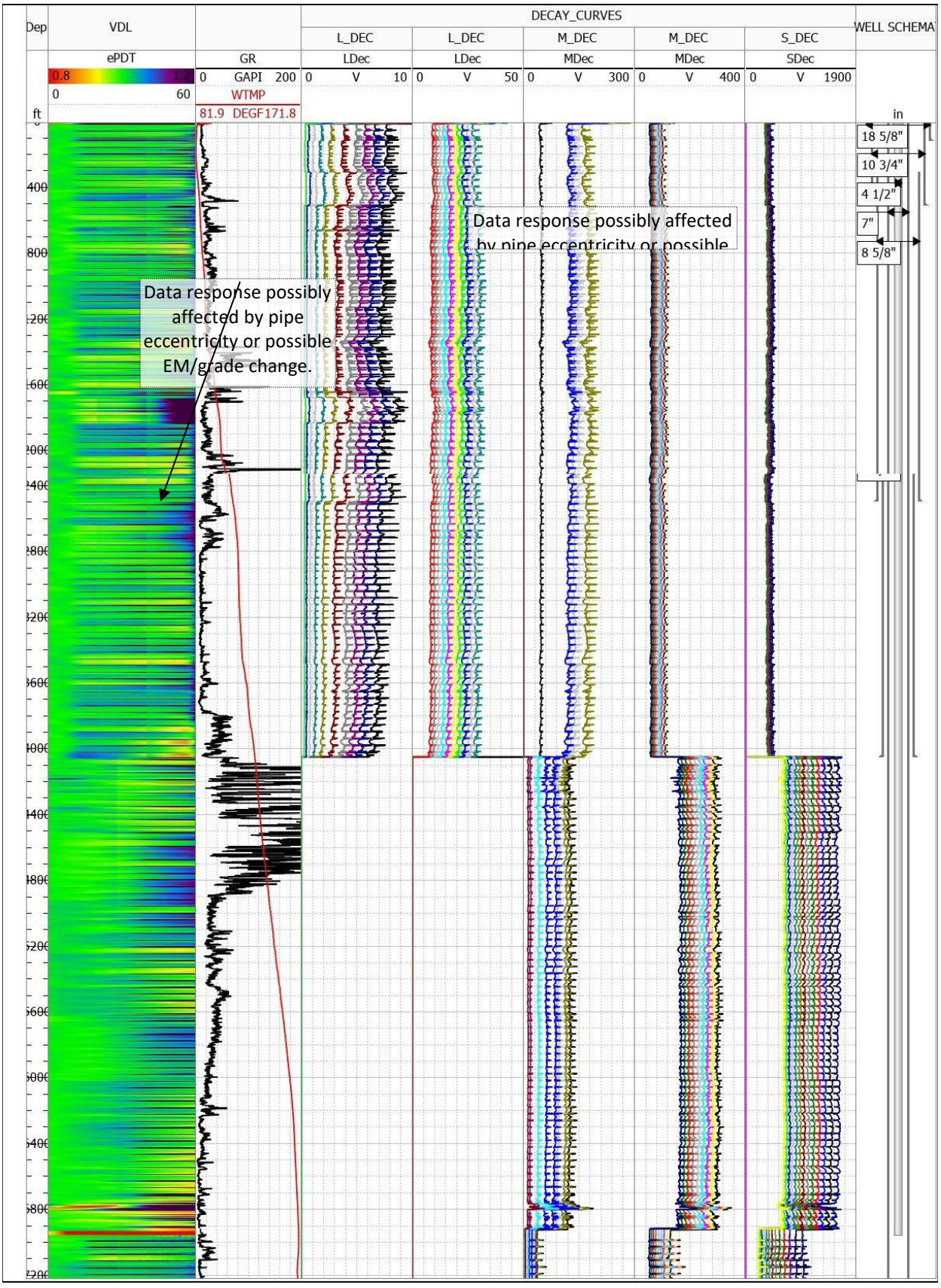


Figure 1 - Raw Data

9. Processed Data Log Display

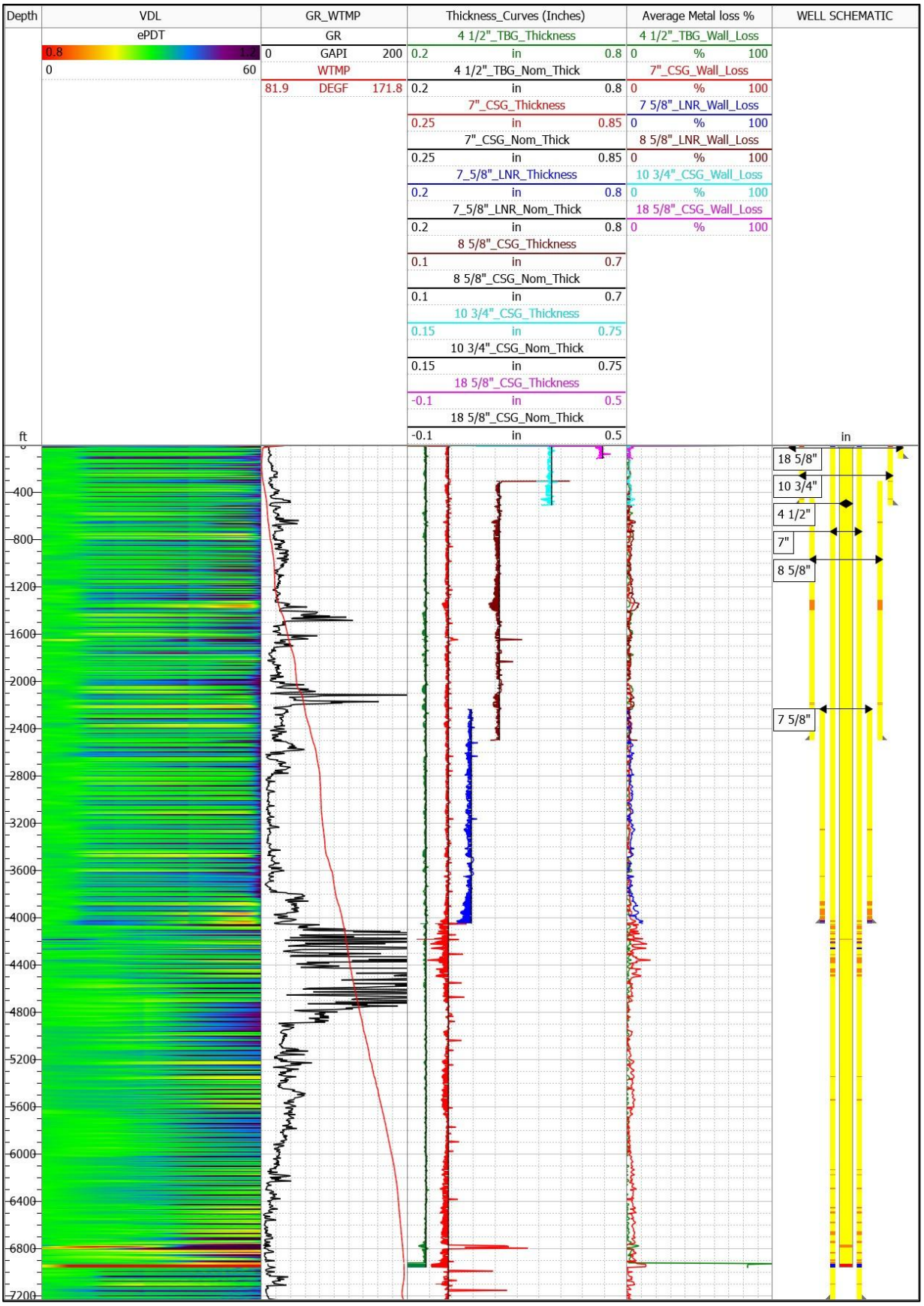


Figure 2 - Four Barriers Metal Loss Evaluation Results

10. Temperature Data and GR Log Display

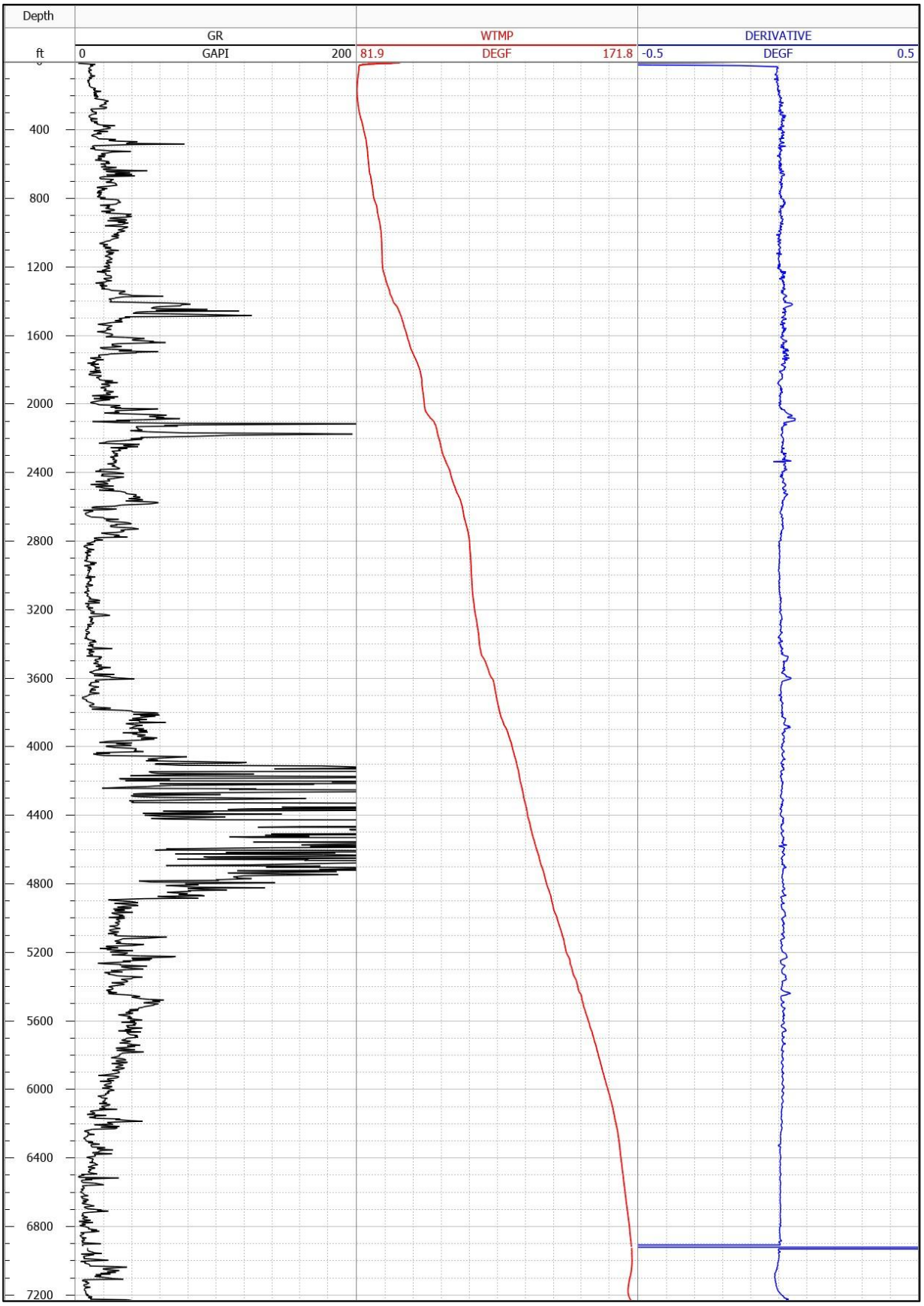


Figure 3 - Expanded Temperature and GR Log

11. Well-head and Hotspots

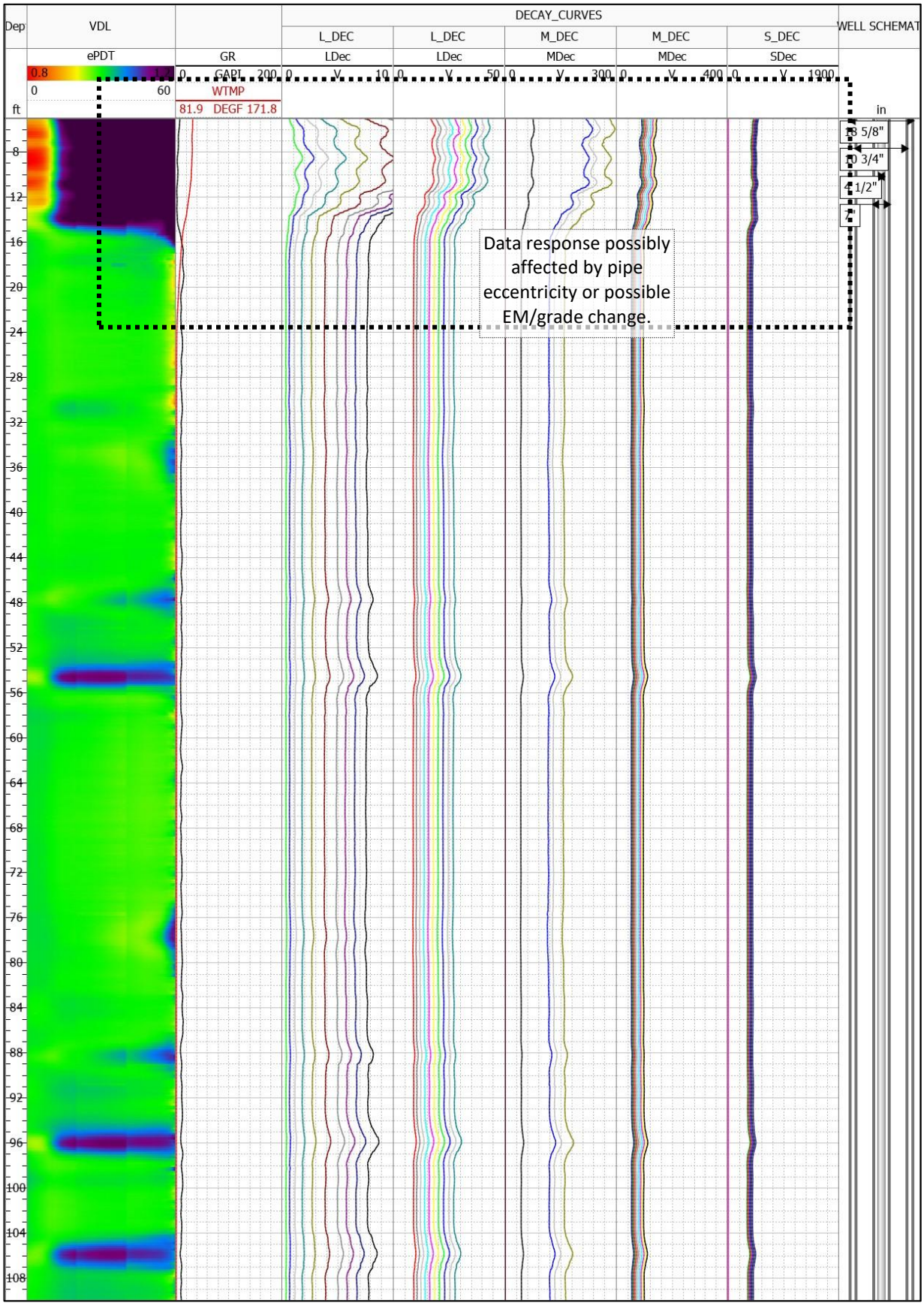


Figure 4 –Well-Head-section

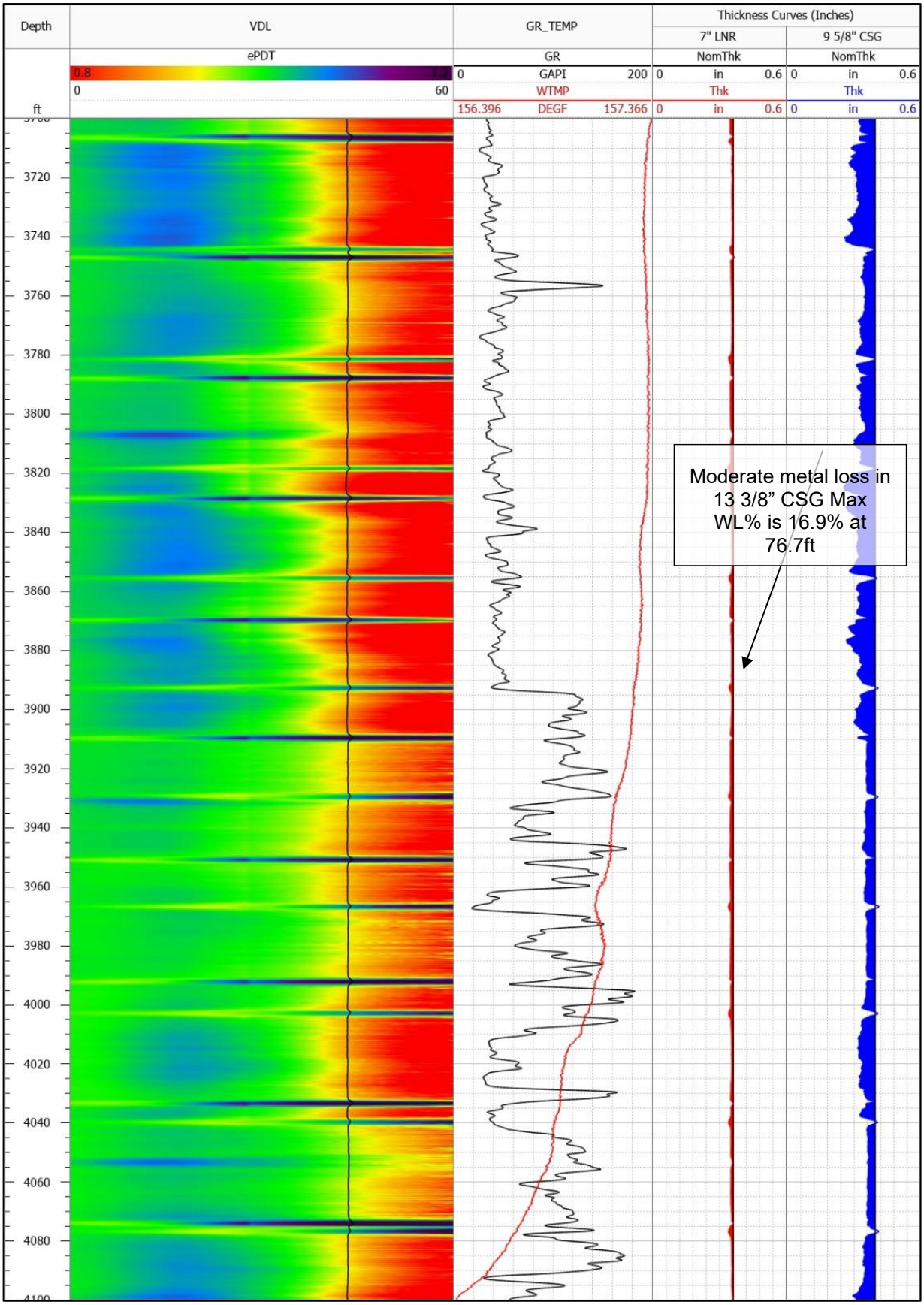


Figure 5 - Highest Metal Loss Channel Selection

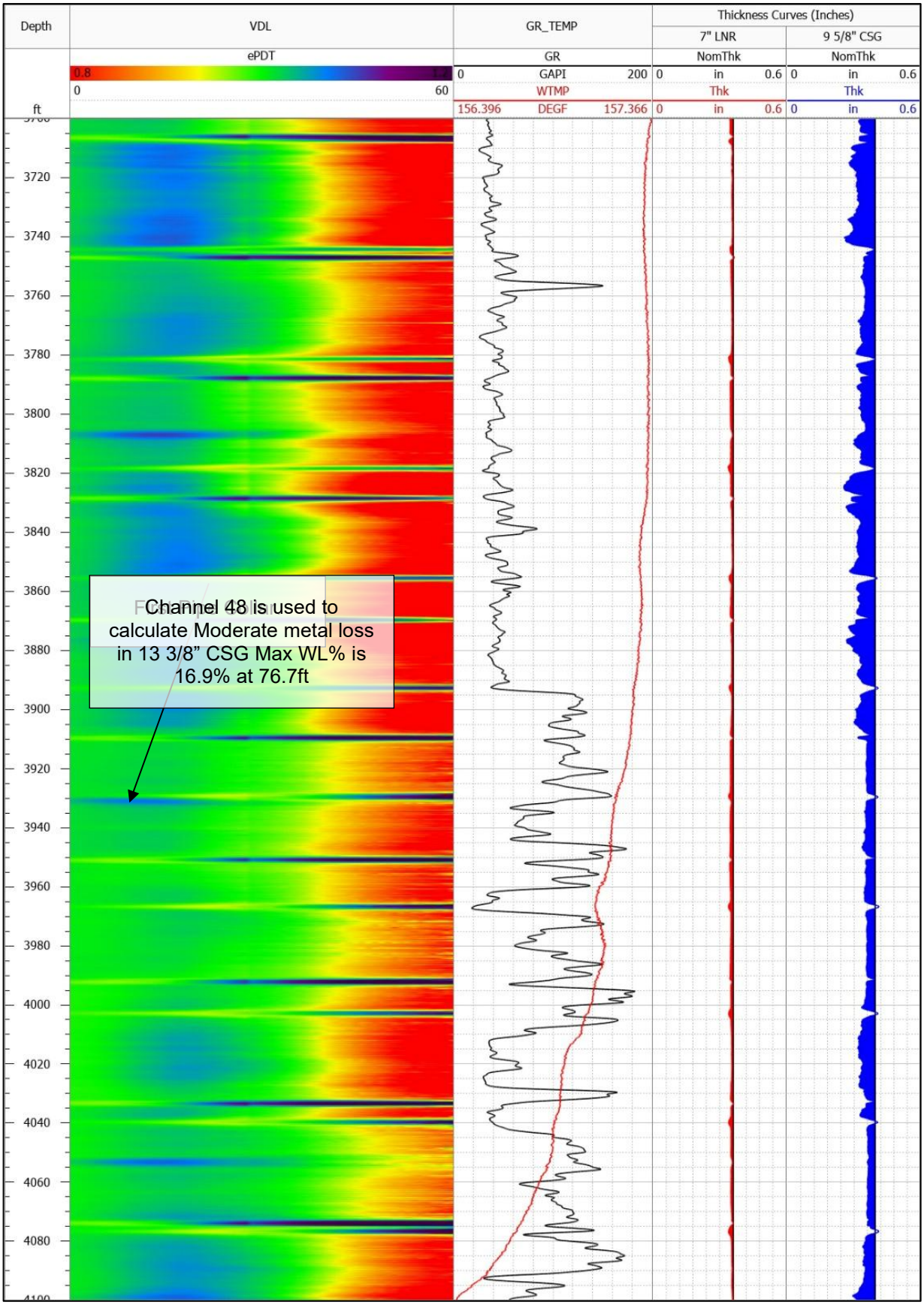
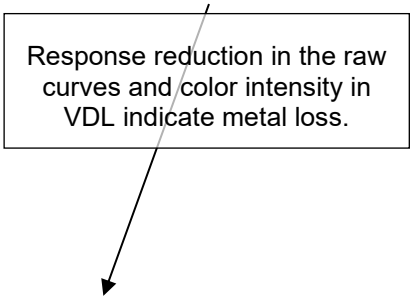


Figure 6 - Highest Metal Loss Channel Selection

{{DMG1 _3}}

Figure 7 - Highest Metal Loss Raw Data



12. Tabulated Joints Metal Loss

Table 10 Joints Metal Loss in 4 1/2" Tubing

#	Top Body (ft)	Bottom Body (ft)	Body Length (ft)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (ft)	Max Loss (%)	Grade	Damage Profile (% wall loss)
1	2909.5	2936.5	27.0	0.362	Affected by Well-head				
2	2938.5	2977.0	38.5	0.362	0.337	2957.7	6.8	B	
3	2979.0	3015.6	36.7	0.362	0.342	2993.9	5.6	B	
4	3017.6	3056.5	38.9	0.362	0.341	3030.9	5.9	B	
5	3058.5	3094.7	36.2	0.362	0.336	3069.8	7.3	B	
6	3096.7	3135.9	39.1	0.362	0.341	3115.2	5.8	B	
7	3137.9	3176.0	38.2	0.362	0.341	3152.4	5.8	B	
8	3178.0	3217.6	39.6	0.362	0.338	3189.1	6.7	B	
9	3219.6	3258.4	38.7	0.362	Affected by 13 3/8" Casing Shoe				
10	3260.4	3299.4	39.0	0.362	0.336	3262.9	7.1	B	
11	3301.4	3339.6	38.2	0.362	0.339	3337.0	6.4	B	
12	3341.6	3379.8	38.2	0.362	0.338	3374.1	6.5	B	
13	3381.8	3419.6	37.9	0.362	0.339	3411.4	6.5	B	
14	3421.6	3461.0	39.3	0.362	0.339	3448.6	6.3	B	
15	3463.0	3502.6	39.6	0.362	0.344	3485.3	4.9	A	
16	3504.6	3543.6	39.0	0.362	0.339	3522.4	6.3	B	
17	3545.6	3584.7	39.0	0.362	0.339	3559.6	6.2	B	
18	3586.7	3623.4	36.7	0.362	0.345	3595.9	4.7	A	
19	3625.4	3664.7	39.3	0.362	0.337	3632.8	6.9	B	
20	3666.7	3705.7	39.0	0.362	0.337	3669.7	6.8	B	
21	3707.7	3746.5	38.7	0.362	0.343	3707.8	5.3	B	
22	3748.5	3787.2	38.7	0.362	0.341	3781.1	5.8	B	
23	3789.2	3827.7	38.5	0.362	0.339	3818.0	6.4	B	
24	3829.7	3869.3	39.6	0.362	0.343	3855.2	5.2	B	
25	3871.3	3908.9	37.6	0.362	0.344	3892.4	5.1	B	
26	3910.9	3950.5	39.6	0.362	0.341	3929.4	5.8	B	
27	3952.5	3991.5	39.0	0.362	0.340	3966.5	6.1	B	
28	3993.5	4032.9	39.3	0.362	0.342	4003.3	5.6	B	
29	4034.9	4072.7	37.9	0.362	0.342	4039.8	5.7	B	
30	4074.7	4113.8	39.0	0.362	0.339	4113.8	6.4	B	
31	4115.8	4155.1	39.3	0.362	0.343	4151.2	5.1	B	
32	4157.1	4195.3	38.2	0.362	0.336	4188.1	7.2	B	
33	4197.3	4236.3	39.0	0.362	0.341	4224.7	5.9	B	
34	4238.3	4277.1	38.7	0.362	0.337	4261.6	7.0	B	
35	4279.1	4318.4	39.3	0.362	0.340	4296.9	6.0	B	
36	4320.4	4359.1	38.7	0.362	0.346	4320.4	4.4	A	
37	4361.1	4400.2	39.0	0.362	0.341	4361.2	5.9	B	
38	4402.2	4440.1	37.9	0.362	0.342	4414.4	5.6	B	
39	4442.1	4480.5	38.4	0.362	0.340	4480.4	6.0	B	
40	4483.3	4521.3	38.0	0.362	0.343	4483.3	5.4	B	
41	4523.3	4562.9	39.6	0.362	0.341	4523.7	5.9	B	
42	4564.9	4603.3	38.5	0.362	0.341	4589.8	5.8	B	

#	Top Body (ft)	Bottom Body (ft)	Body Length (ft)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (ft)	Max Loss (%)	Grade	Damage Profile (% wall loss)
43	4605.3	4643.8	38.5	0.362	0.347	4642.9	4.1	A	
44	4645.8	4684.8	39.0	0.362	0.342	4662.9	5.4	B	
45	4686.8	4725.3	38.5	0.362	0.343	4687.1	5.2	B	
46	4727.3	4766.6	39.3	0.362	0.337	4737.7	6.9	B	
47	4768.6	4807.4	38.7	0.362	0.335	4774.7	7.6	B	
48	4809.4	4848.4	39.0	0.362	0.332	4848.3	8.3	B	
49	4850.4	4883.6	33.2	0.362	0.345	4883.6	4.6	A	
50	4885.6	4908.1	22.4	0.362	ECP				
51	4910.1	4952.0	41.9	0.362	0.337	4922.7	7.0	B	
52	4954.0	4990.1	36.1	0.362	0.329	4963.3	9.2	B	
53	4992.1	5031.2	39.0	0.362	0.344	4999.6	4.9	A	
54	5033.2	5075.1	41.9	0.362	0.334	5075.1	7.7	B	
55	5077.1	5110.9	33.8	0.362	Affected by 9 5/8" Casing Shoe				
56	5112.9	5151.7	38.7	0.362	0.337	5136.3	6.8	B	
57	5153.7	5192.4	38.7	0.362	0.346	5188.4	4.4	A	
58	5194.4	5232.3	37.9	0.362	0.344	5229.8	4.9	A	
59	5234.3	5272.5	38.2	0.362	0.347	5263.7	4.2	A	
60	5274.5	5312.6	38.2	0.362	0.343	5310.0	5.3	B	
61	5314.6	5353.4	38.7	0.362	0.343	5350.8	5.2	B	
62	5355.4	5393.5	38.2	0.362	0.341	5389.1	5.7	B	
63	5395.5	5434.6	39.0	0.362	0.342	5402.1	5.4	B	
64	5436.6	5475.9	39.3	0.362	0.337	5441.8	6.9	B	
65	5477.9	5516.4	38.5	0.362	0.339	5512.8	6.2	B	
66	5518.4	5557.1	38.7	0.362	0.336	5553.8	7.1	B	
67	5559.1	5597.3	38.2	0.362	0.338	5584.2	6.6	B	
68	5599.3	5638.3	39.0	0.362	0.333	5632.3	8.0	B	
69	5640.3	5679.1	38.7	0.362	0.337	5644.8	6.8	B	
70	5681.1	5719.5	38.5	0.362	0.338	5683.7	6.7	B	
71	5721.5	5760.8	39.3	0.362	0.339	5757.9	6.3	B	
72	5762.8	5801.9	39.0	0.362	0.336	5769.9	7.2	B	
73	5803.9	5842.0	38.2	0.362	0.331	5836.3	8.6	B	
74	5844.0	5882.8	38.7	0.362	0.333	5849.0	7.9	B	
75	5884.8	5923.3	38.5	0.362	0.348	5921.4	4.0	A	
76	5925.3	5964.0	38.7	0.362	0.340	5933.8	6.1	B	
77	5966.0	6003.0	37.0	0.362	0.343	5998.2	5.2	B	
78	6005.0	6041.1	36.1	0.362	0.343	6011.0	5.1	B	
79	6043.1	6082.2	39.0	0.362	0.341	6068.1	5.9	B	
80	6084.2	6121.2	37.0	0.362	0.340	6087.1	6.2	B	
81	6123.2	6127.0	3.8	0.362	0.349	6125.8	3.6	A	

Table 11 Joints Metal Loss in 7" Casing

#	Top Body (ft)	Bottom Body (ft)	Body Length (ft)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (ft)	Max Loss (%)	Grade	Damage Profile (% wall loss)
1	30.0	51.3	21.3	0.395	Affected by Well-head				
2	53.3	86.8	33.5	0.395	0.376	54.8	4.9	A	
3	88.8	124.9	36.1	0.395	0.378	90.3	4.4	A	
4	126.9	161.9	35.0	0.395	0.378	131.6	4.3	A	
5	163.9	198.8	35.0	0.395	0.380	171.5	3.8	A	
6	200.8	236.1	35.3	0.395	0.380	213.1	3.8	A	
7	238.1	273.1	35.0	0.395	0.381	238.7	3.6	A	
8	275.1	309.7	34.7	0.352	0.336	304.6	4.5	A	
9	311.7	349.0	37.3	0.352	0.340	333.5	3.5	A	
10	351.0	388.0	37.0	0.352	0.339	373.8	3.8	A	
11	390.0	425.6	35.5	0.352	0.332	413.8	5.6	B	
12	427.6	462.8	35.3	0.352	0.332	455.7	5.7	B	
13	464.8	498.9	34.1	0.352	0.334	496.7	5.2	B	
14	500.9	537.3	36.4	0.352	0.333	535.4	5.4	B	
15	539.3	573.7	34.4	0.352	0.331	543.7	5.8	B	
16	575.7	610.7	35.0	0.352	0.321	601.7	8.9	B	

#	Top Body (ft)	Bottom Body (ft)	Body Length (ft)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (ft)	Max Loss (%)	Grade	Damage Profile (% wall loss)
17	612.7	649.7	37.0	0.352	0.331	630.7	5.9	B	
18	651.7	687.5	35.8	0.352	0.329	667.3	6.5	B	
19	689.5	725.1	35.5	0.352	0.333	710.9	5.4	B	
20	727.1	764.4	37.3	0.352	0.327	747.0	7.2	B	
21	766.4	801.0	34.7	0.352	0.335	799.6	4.7	A	
22	803.0	838.9	35.8	0.352	0.336	832.7	4.6	A	
23	840.9	875.5	34.7	0.352	0.336	858.2	4.6	A	
24	877.5	912.8	35.3	0.352	0.340	886.2	3.4	A	
25	914.8	950.6	35.8	0.352	0.337	936.7	4.3	A	
26	952.6	987.0	34.4	0.352	0.333	970.3	5.3	B	
27	989.0	1023.7	34.7	0.352	0.337	996.5	4.3	A	
28	1025.7	1061.8	36.1	0.352	0.333	1030.1	5.5	B	
29	1063.8	1098.2	34.4	0.352	0.333	1077.3	5.3	B	
30	1100.2	1136.3	36.1	0.352	0.329	1116.9	6.6	B	
31	1138.3	1173.0	34.7	0.352	0.332	1154.2	5.8	B	
32	1175.0	1209.1	34.1	0.352	0.332	1197.7	5.6	B	
33	1211.1	1247.8	36.7	0.352	0.331	1212.5	6.0	B	
34	1249.8	1284.2	34.4	0.352	0.331	1280.5	6.0	B	
35	1286.2	1321.1	35.0	0.352	0.328	1303.0	6.9	B	
36	1323.1	1358.4	35.3	0.352	0.332	1343.8	5.7	B	
37	1360.4	1395.4	35.0	0.352	0.334	1361.2	5.2	B	
38	1397.4	1432.9	35.5	0.352	0.327	1414.2	7.2	B	
39	1434.9	1470.2	35.3	0.352	0.329	1439.6	6.4	B	
40	1472.2	1507.1	35.0	0.352	0.331	1483.8	5.9	B	
41	1509.1	1544.1	35.0	0.352	0.327	1525.5	7.2	B	
42	1546.1	1581.1	35.0	0.352	0.331	1558.7	6.1	B	
43	1583.1	1618.9	35.8	0.352	0.334	1609.2	5.2	B	
44	1620.9	1656.4	35.5	0.352	0.329	1640.2	6.6	B	
45	1658.4	1693.7	35.3	0.352	0.335	1671.7	4.9	A	
46	1695.7	1730.4	34.7	0.352	0.337	1723.4	4.2	A	
47	1732.4	1767.6	35.3	0.352	0.340	1748.2	3.4	A	
48	1769.6	1806.6	37.0	0.352	0.331	1800.0	5.9	B	
49	1808.6	1843.6	35.0	0.352	0.339	1837.8	3.7	A	
50	1845.6	1881.1	35.5	0.352	0.337	1872.9	4.3	A	
51	1883.1	1917.5	34.4	0.352	0.334	1886.1	5.1	B	
52	1919.5	1954.2	34.7	0.352	0.336	1952.7	4.5	A	
53	1956.2	1992.3	36.1	0.352	0.335	1973.6	4.9	A	
54	1994.3	2029.0	34.7	0.352	0.330	1994.4	6.2	B	
55	2031.0	2066.5	35.5	0.352	0.333	2046.4	5.4	B	
56	2068.5	2105.2	36.7	0.352	0.329	2073.3	6.6	B	
57	2107.2	2142.5	35.3	0.352	0.336	2109.6	4.5	A	
58	2144.5	2179.7	35.3	0.395	0.365	2156.7	7.7	B	
59	2181.7	2216.4	34.7	0.395	0.375	2196.1	5.2	B	
60	2218.4	2253.1	34.7	0.395	0.379	2228.2	4.1	A	
61	2255.1	2290.3	35.3	0.395	0.375	2278.1	5.2	B	

#	Top Body (ft)	Bottom Body (ft)	Body Length (ft)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (ft)	Max Loss (%)	Grade	Damage Profile (% wall loss)
62	2292.3	2327.3	35.0	0.395	0.372	2299.7	5.7	B	
63	2329.3	2364.9	35.5	0.395	0.379	2330.2	3.9	A	
64	2366.9	2401.8	35.0	0.395	0.371	2389.9	6.1	B	
65	2403.8	2439.1	35.3	0.395	0.374	2412.7	5.2	B	
66	2441.1	2475.7	34.7	0.395	0.380	2464.8	3.7	A	
67	2477.7	2513.0	35.3	0.395	0.375	2511.9	5.2	B	
68	2515.0	2549.7	34.7	0.395	0.365	2538.7	7.6	B	
69	2551.7	2586.6	35.0	0.395	0.375	2552.5	5.0	B	
70	2588.6	2623.9	35.3	0.395	0.374	2604.6	5.2	B	
71	2625.9	2661.1	35.3	0.395	0.378	2634.2	4.2	A	
72	2663.1	2698.4	35.3	0.395	0.386	2674.2	2.3	A	
73	2700.4	2735.1	34.7	0.395	0.372	2716.7	5.8	B	
74	2737.1	2771.7	34.7	0.395	0.365	2748.2	7.6	B	
75	2773.7	2808.7	35.0	0.395	0.377	2790.9	4.6	A	
76	2810.7	2845.7	35.0	0.395	0.366	2834.5	7.2	B	
77	2847.7	2882.9	35.3	0.395	0.375	2869.7	5.1	B	
78	2884.9	2919.9	35.0	0.395	Affected by 7" Liner Hanger				
79	2921.9	2956.6	34.7	0.395	0.374	2940.7	5.4	B	
80	2958.6	2993.2	34.7	0.395	0.379	2965.5	4.1	A	
81	2995.2	3030.2	35.0	0.395	0.379	2996.4	4.1	A	
82	3032.2	3066.0	33.8	0.395	0.377	3065.9	4.6	A	
83	3068.0	3077.9	9.9	0.395	DVP				
84	3079.9	3114.3	34.4	0.395	0.375	3089.7	5.2	B	
85	3116.3	3151.0	34.7	0.395	0.377	3117.0	4.6	A	
86	3153.0	3188.5	35.5	0.395	0.378	3155.4	4.4	A	
87	3190.5	3225.2	34.7	0.395	0.378	3195.4	4.3	A	
88	3227.2	3262.2	35.0	0.395	Affected by 13 3/8" Casing Shoe				
89	3264.2	3299.7	35.5	0.395	0.372	3294.4	5.9	B	
90	3301.7	3336.4	34.7	0.395	0.369	3317.9	6.7	B	
91	3338.4	3373.3	35.0	0.395	0.372	3343.3	5.9	B	
92	3375.3	3410.6	35.3	0.395	0.374	3377.4	5.4	B	
93	3412.6	3448.1	35.5	0.395	0.365	3439.1	7.6	B	
94	3450.1	3484.5	34.4	0.395	0.375	3483.5	5.0	B	
95	3486.5	3521.5	35.0	0.395	0.364	3487.9	7.7	B	
96	3523.5	3559.3	35.8	0.395	0.346	3534.9	12.3	C	
97	3561.3	3594.8	33.5	0.395	0.292	3579.1	25.9	D	
98	3596.8	3632.4	35.5	0.395	0.356	3605.1	9.9	B	
99	3634.4	3669.3	35.0	0.395	0.303	3662.2	23.2	D	
100	3671.3	3705.7	34.4	0.395	0.320	3700.9	18.9	C	
101	3707.7	3743.9	36.1	0.395	0.255	3740.4	35.4	D	
102	3745.9	3780.5	34.7	0.395	0.308	3779.6	22.0	D	
103	3782.5	3817.5	35.0	0.395	0.297	3810.2	24.8	D	
104	3819.5	3855.0	35.5	0.395	0.253	3824.9	35.9	D	
105	3857.0	3892.3	35.3	0.395	0.266	3876.8	32.8	D	
106	3894.3	3928.7	34.4	0.395	0.299	3904.4	24.4	D	

#	Top Body (ft)	Bottom Body (ft)	Body Length (ft)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (ft)	Max Loss (%)	Grade	Damage Profile (% wall loss)
107	3930.7	3966.2	35.5	0.395	0.336	3946.5	15.0	C	
108	3968.2	4002.0	33.8	0.395	0.335	4000.9	15.3	C	
109	4004.0	4039.0	35.0	0.395	0.317	4014.7	19.8	C	
110	4041.0	4075.9	35.0	0.395	0.357	4070.4	9.7	B	
111	4077.9	4113.2	35.3	0.395	0.293	4098.3	25.7	D	
112	4115.2	4150.2	35.0	0.395	0.303	4117.6	23.3	D	
113	4152.2	4187.4	35.3	0.395	0.344	4183.5	12.8	C	
114	4189.4	4223.5	34.1	0.395	0.363	4201.9	8.1	B	
115	4225.5	4260.5	35.0	0.395	0.371	4231.6	6.0	B	
116	4262.5	4296.0	33.5	0.395	0.376	4283.0	4.7	A	
117	4298.0	4333.2	35.3	0.395	0.370	4305.8	6.4	B	
118	4335.2	4370.5	35.3	0.395	0.381	4357.4	3.5	A	
119	4372.5	4407.5	35.0	0.395	0.378	4406.4	4.2	A	
120	4409.5	4444.1	34.7	0.395	0.383	4430.3	3.0	A	
121	4446.1	4480.8	34.7	0.395	0.384	4478.5	2.9	A	
122	4482.8	4517.5	34.7	0.395	0.382	4485.5	3.3	A	
123	4519.5	4553.6	34.1	0.395	0.380	4547.8	3.7	A	
124	4555.6	4589.9	34.4	0.395	0.382	4567.5	3.3	A	
125	4591.9	4627.5	35.5	0.395	0.385	4613.4	2.5	A	
126	4629.5	4663.6	34.1	0.395	0.380	4646.8	3.9	A	
127	4665.6	4699.1	33.5	0.395	0.384	4689.7	2.7	A	
128	4701.1	4736.6	35.5	0.395	0.384	4729.7	2.7	A	
129	4738.6	4774.2	35.5	0.395	0.382	4771.4	3.2	A	
130	4776.2	4811.1	35.0	0.395	0.381	4777.8	3.5	A	
131	4813.1	4848.1	35.0	0.395	0.386	4814.4	2.3	A	
132	4850.1	4884.2	34.1	0.395	0.375	4863.7	4.9	A	
133	4886.2	4924.9	38.7	0.395	0.366	4901.3	7.3	B	
134	4926.9	4962.5	35.5	0.395	0.376	4928.8	4.8	A	
135	4964.5	4999.5	35.0	0.395	0.382	4995.4	3.4	A	
136	5001.5	5038.2	36.7	0.395	0.386	5022.3	2.3	A	
137	5040.2	5075.5	35.3	0.395	0.378	5050.1	4.2	A	

Table 12 Joints Metal Loss in 7 5/8" Liner

#	Top Body (ft)	Bottom Body (ft)	Body Length (ft)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (ft)	Max Loss (%)	Grade	Damage Profile (% wall loss)
1	30.0	50.8	20.8	0.430	Affected by Well-head				
2	52.8	88.0	35.2	0.430	0.404	56.0	6.1	B	
3	90.0	131.3	41.4	0.430	0.403	114.2	6.3	B	
4	133.3	172.1	38.7	0.430	0.406	137.6	5.5	B	
5	174.1	212.5	38.5	0.430	0.410	178.2	4.7	A	
6	214.5	252.7	38.2	0.430	0.404	251.5	6.1	B	
7	254.7	293.4	38.7	0.430	0.404	261.5	6.1	B	
8	295.4	332.7	37.3	0.430	0.404	314.0	6.0	B	
9	334.7	372.9	38.2	0.430	0.407	335.5	5.3	B	
10	374.9	413.4	38.5	0.430	0.402	381.5	6.4	B	
11	415.4	454.1	38.7	0.430	0.416	419.8	3.3	A	
12	456.1	494.6	38.5	0.430	0.405	490.8	5.9	B	
13	496.6	533.3	36.7	0.430	0.401	519.1	6.7	B	
14	535.3	545.5	10.2	0.430	DVP				
15	547.5	586.5	39.0	0.430	0.401	555.4	6.8	B	
16	588.5	626.4	37.9	0.430	0.401	617.7	6.7	B	
17	628.4	667.7	39.3	0.430	0.403	638.6	6.3	B	
18	669.7	709.1	39.3	0.430	0.405	675.5	5.7	B	
19	711.1	749.8	38.7	0.430	0.404	731.9	6.2	B	
20	751.8	791.1	39.3	0.430	0.403	758.9	6.4	B	
21	793.1	831.0	37.9	0.430	0.413	809.2	3.9	A	
22	833.0	871.2	38.2	0.430	0.408	850.9	5.1	B	
23	873.2	913.1	39.9	0.430	0.412	884.2	4.2	A	
24	915.1	953.5	38.5	0.430	0.417	938.7	2.9	A	
25	955.5	995.2	39.6	0.430	0.413	963.1	4.0	A	
26	997.2	1035.3	38.2	0.430	0.415	1000.9	3.5	A	

#	Top Body (ft)	Bottom Body (ft)	Body Length (ft)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (ft)	Max Loss (%)	Grade	Damage Profile (% wall loss)
27	1037.3	1075.2	37.9	0.430	0.411	1068.1	4.5	A	
28	1077.2	1116.8	39.6	0.430	0.413	1095.1	3.9	A	
29	1118.8	1154.4	35.5	0.430	0.403	1151.9	6.4	B	
30	1156.4	1195.7	39.3	0.430	0.404	1169.9	6.0	B	
31	1197.7	1237.3	39.6	0.430	0.403	1231.1	6.2	B	
32	1239.3	1278.6	39.3	0.430	0.402	1244.6	6.4	B	
33	1280.6	1320.0	39.3	0.430	0.407	1297.5	5.3	B	
34	1322.0	1358.4	36.4	0.430	0.404	1346.2	6.0	B	
35	1360.4	1399.1	38.7	0.430	0.397	1372.6	7.7	B	
36	1401.1	1439.6	38.5	0.430	0.405	1406.2	5.9	B	
37	1441.6	1480.3	38.7	0.430	0.402	1455.6	6.5	B	
38	1482.3	1519.6	37.3	0.430	0.400	1516.2	7.0	B	
39	1521.6	1558.9	37.3	0.430	0.401	1537.1	6.7	B	
40	1560.9	1599.1	38.2	0.430	0.415	1566.8	3.6	A	
41	1601.1	1639.0	37.9	0.430	0.404	1629.3	6.1	B	
42	1641.0	1680.6	39.6	0.430	0.405	1678.9	5.9	B	
43	1682.6	1717.6	35.0	0.430	0.401	1709.6	6.7	B	
44	1719.6	1757.1	37.6	0.430	0.400	1722.8	6.9	B	
45	1759.1	1797.9	38.7	0.430	0.393	1786.2	8.6	B	
46	1799.9	1838.1	38.2	0.430	0.393	1804.3	8.5	B	
47	1840.1	1873.0	32.9	0.430	0.404	1841.9	6.1	B	
48	1875.0	1913.1	38.2	0.430	0.409	1905.2	4.9	A	
49	1915.1	1954.2	39.0	0.430	0.402	1926.5	6.5	B	
50	1956.2	1992.3	36.1	0.430	0.400	1976.8	6.9	B	
51	1994.3	2033.9	39.6	0.430	0.402	2032.7	6.6	B	
52	2035.9	2073.8	37.9	0.430	0.391	2070.7	9.2	B	
53	2075.8	2114.8	39.0	0.430	0.388	2076.9	9.7	B	
54	2116.8	2155.0	38.2	0.430	0.394	2128.9	8.5	B	
55	2157.0	2196.3	39.3	0.430	0.410	2168.2	4.7	A	
56	2198.3	2237.7	39.3	0.430	0.411	2204.5	4.5	A	
57	2239.7	2276.7	37.0	0.430	0.407	2262.6	5.3	B	
58	2278.7	2314.5	35.8	0.430	0.409	2301.3	4.9	A	
59	2316.5	2353.8	37.3	0.430	0.409	2334.7	4.9	A	
60	2355.8	2389.9	34.1	0.430	0.405	2376.9	5.8	B	
61	2391.9	2431.2	39.3	0.430	0.406	2426.2	5.7	B	
62	2433.2	2471.7	38.5	0.430	0.402	2452.6	6.4	B	
63	2473.7	2512.7	39.0	0.430	0.407	2499.6	5.3	B	
64	2514.7	2552.9	38.2	0.430	0.398	2543.4	7.4	B	
65	2554.9	2593.3	38.5	0.430	0.402	2577.0	6.5	B	
66	2595.3	2634.1	38.7	0.430	0.406	2629.1	5.5	B	
67	2636.1	2674.2	38.2	0.430	0.403	2637.2	6.3	B	
68	2676.2	2714.4	38.2	0.430	0.406	2683.7	5.7	B	
69	2716.4	2753.4	37.0	0.430	0.398	2747.2	7.4	B	
70	2755.4	2791.5	36.1	0.430	0.400	2763.5	7.0	B	
71	2793.5	2831.1	37.6	0.430	0.400	2817.7	7.0	B	

#	Top Body (ft)	Bottom Body (ft)	Body Length (ft)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (ft)	Max Loss (%)	Grade	Damage Profile (% wall loss)
72	2833.1	2871.9	38.7	0.430	0.397	2856.8	7.7	B	
73	2873.9	2915.8	41.9	0.430	Affected by 7" Liner Hanger				
74	2917.8	2955.4	37.6	0.430	0.401	2929.7	6.8	B	
75	2957.4	2993.5	36.1	0.430	0.404	2982.8	6.0	B	
76	2995.5	3035.4	39.9	0.430	0.399	3032.9	7.1	B	
77	3037.4	3073.6	36.1	0.430	0.401	3060.2	6.8	B	
78	3075.6	3111.1	35.5	0.430	0.402	3102.2	6.5	B	
79	3113.1	3151.9	38.7	0.430	0.400	3132.6	6.9	B	
80	3153.9	3192.9	39.0	0.430	0.407	3183.9	5.3	B	
81	3194.9	3234.2	39.3	0.430	0.408	3212.2	5.1	B	

Table 13 Joints Metal Loss in 8 5/8" Liner

#	Top Body (ft)	Bottom Body (ft)	Body Length (ft)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (ft)	Max Loss (%)	Grade	Damage Profile (% wall loss)
1	30.0	54.2	24.2	0.435	Affected by Well-head				
2	56.2	97.3	41.1	0.435	0.417	60.7	4.1	A	
3	99.3	139.2	39.9	0.435	0.416	135.7	4.4	A	
4	141.2	178.8	37.6	0.435	0.416	175.6	4.4	A	
5	180.8	221.0	40.2	0.435	0.417	208.1	4.1	A	
6	223.0	262.3	39.3	0.435	0.410	249.3	5.7	B	
7	264.3	304.2	39.9	0.435	0.414	267.0	4.9	A	
8	306.2	346.7	40.5	0.435	0.393	319.2	9.6	B	
9	348.7	388.0	39.3	0.435	0.395	382.4	9.2	B	
10	390.0	430.2	40.2	0.435	0.410	407.6	5.8	B	
11	432.2	472.7	40.5	0.435	0.406	468.7	6.6	B	
12	474.7	514.1	39.3	0.435	0.398	510.2	8.4	B	
13	516.1	556.6	40.5	0.435	0.401	553.6	7.9	B	
14	558.6	601.5	42.9	0.435	0.403	595.9	7.3	B	

Table 14 Joints Metal Loss in 10 3/4" Casing

#	Top Body (ft)	Bottom Body (ft)	Body Length (ft)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (ft)	Max Loss (%)	Grade	Damage Profile (% wall loss)
1	30.0	70.2	40.2	0.688	Affected by Well-head				
2	72.2	114.0	41.8	0.688	0.635	74.6	7.7	B	

Table 15 Joints Metal Loss in 18 5/8" Casing

#	Top Body (ft)	Bottom Body (ft)	Body Length (ft)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (ft)	Max Loss (%)	Grade	Damage Profile (% wall loss)
1	35.0	53.3	18.3	0.250	Affected by Well-head				
2	55.3	89.8	34.5	0.250	0.241	62.5	3.5	A	

#	Top Body (ft)	Bottom Body (ft)	Body Length (ft)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (ft)	Max Loss (%)	Grade	Damage Profile (% wall loss)
3	91.8	98.2	6.4	0.250	0.256	93.9	-2.5	E	
4	100.2	106.6	6.4	0.250	0.237	104.0	5.3	B	
5	108.6	147.3	38.7	0.250	0.241	144.4	3.6	A	
6	149.3	187.8	38.4	0.250	0.243	185.1	2.9	A	
7	189.8	228.2	38.4	0.250	0.242	225.1	3.0	A	
8	230.2	268.9	38.7	0.250	0.241	264.5	3.8	A	
9	270.9	308.8	37.9	0.250	0.241	304.2	3.5	A	
10	310.8	347.6	36.8	0.250	0.238	345.1	4.8	A	
11	349.6	388.1	38.4	0.250	0.238	385.9	4.9	A	
12	390.1	428.8	38.7	0.250	0.239	426.6	4.3	A	
13	430.8	469.5	38.7	0.250	0.246	465.2	1.5	A	
14	471.5	509.1	37.6	0.250	0.242	504.2	3.4	A	
15	511.1	549.6	38.4	0.250	0.237	546.6	5.0	B	
16	551.6	590.0	38.4	0.250	0.239	587.6	4.3	A	
17	592.0	631.0	39.0	0.250	0.238	626.8	4.7	A	
18	633.0	670.6	37.6	0.250	0.240	667.4	3.9	A	
19	672.6	711.1	38.4	0.250	0.238	707.4	4.8	A	
20	713.1	751.8	38.7	0.250	0.239	745.4	4.3	A	
21	753.8	793.6	39.8	0.250	0.239	787.3	4.3	A	
22	795.6	834.6	39.0	0.250	0.240	827.3	4.1	A	
23	836.6	876.1	39.5	0.250	0.239	865.7	4.3	A	
24	878.1	916.3	38.2	0.250	0.241	905.0	3.8	A	
25	918.3	957.0	38.7	0.250	0.239	944.4	4.5	A	
26	959.0	998.8	39.8	0.250	0.240	963.1	4.2	A	
27	1000.8	1038.9	38.2	0.250	0.236	1002.0	5.4	B	
28	1040.9	1079.9	39.0	0.250	0.241	1078.6	3.6	A	
29	1081.9	1120.1	38.2	0.250	0.242	1119.0	3.1	A	
30	1122.1	1160.8	38.7	0.250	0.243	1146.6	2.7	A	
31	1162.8	1201.0	38.2	0.250	0.234	1189.1	6.2	B	
32	1203.0	1241.4	38.4	0.250	0.237	1204.6	5.1	B	
33	1243.4	1281.6	38.2	0.250	0.242	1277.6	3.1	A	
34	1283.6	1323.1	39.5	0.250	0.241	1307.3	3.7	A	
35	1325.1	1363.5	38.4	0.250	0.241	1362.3	3.4	A	
36	1365.5	1403.7	38.2	0.250	0.238	1402.6	4.7	A	
37	1405.7	1444.7	39.0	0.250	0.242	1431.1	3.1	A	
38	1446.7	1485.1	38.4	0.250	0.241	1474.1	3.6	A	
39	1487.1	1525.6	38.4	0.250	0.244	1488.0	2.6	A	
40	1527.6	1567.1	39.5	0.250	0.239	1551.5	4.2	A	
41	1569.1	1606.5	37.4	0.250	0.243	1590.1	2.7	A	
42	1608.5	1646.6	38.2	0.250	0.242	1630.0	3.3	A	
43	1648.6	1687.9	39.3	0.250	0.240	1668.5	3.8	A	
44	1689.9	1728.6	38.7	0.250	0.241	1713.2	3.8	A	
45	1730.6	1769.9	39.3	0.250	Affected by liner				
46	1771.9	1810.3	38.4	0.250	Affected by liner				
47	1812.3	1849.9	37.6	0.250	0.242	1848.9	3.3	A	

#	Top Body (ft)	Bottom Body (ft)	Body Length (ft)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (ft)	Max Loss (%)	Grade	Damage Profile (% wall loss)
48	1851.9	1891.4	39.5	0.250	0.238	1873.0	4.9	A	
49	1893.4	1931.9	38.4	0.250	0.237	1917.9	5.1	B	
50	1933.9	1972.3	38.4	0.250	0.241	1958.3	3.4	A	
51	1974.3	2012.8	38.4	0.250	0.240	1975.9	4.2	A	
52	2014.8	2054.3	39.5	0.250	0.239	2016.5	4.3	A	
53	2056.3	2095.5	39.3	0.250	affected by liner				
54	2097.5	2136.8	39.3	0.250	0.243	2120.0	2.8	A	
55	2138.8	2178.1	39.3	0.250	0.248	2140.1	0.9	A	
56	2180.1	2219.3	39.3	0.250	0.246	2204.7	1.8	A	
57	2221.3	2260.3	39.0	0.250	0.245	2241.4	1.9	A	
58	2262.3	2301.6	39.3	0.250	0.245	2286.7	2.0	A	
59	2303.6	2343.1	39.5	0.250	0.243	2327.3	2.6	A	
60	2345.1	2384.3	39.3	0.250	0.245	2364.6	2.0	A	
61	2386.3	2423.7	37.4	0.250	0.242	2394.8	3.3	A	
62	2425.7	2464.1	38.4	0.250	0.243	2426.4	2.7	A	
63	2466.1	2504.8	38.7	0.250	0.246	2487.7	1.6	A	
64	2506.8	2544.7	37.9	0.250	0.242	2520.2	3.3	A	
65	2546.7	2586.0	39.3	0.250	0.238	2567.2	4.7	A	
66	2588.0	2626.4	38.4	0.250	0.240	2625.2	4.2	A	
67	2628.4	2664.4	36.0	0.250	0.244	2641.2	2.5	A	
68	2666.4	2705.7	39.3	0.250	0.238	2681.1	4.6	A	
69	2707.7	2746.7	39.0	0.250	0.241	2732.8	3.8	A	
70	2748.7	2787.1	38.4	0.250	0.240	2773.7	3.9	A	
71	2789.1	2828.4	39.3	0.250	0.240	2815.0	3.9	A	
72	2830.4	2869.1	38.7	0.250	0.243	2855.8	3.0	A	
73	2871.1	2909.8	38.7	0.250	0.240	2896.7	4.2	A	
74	2911.8	2950.2	38.4	0.250	0.244	2936.9	2.4	A	
75	2952.2	2990.1	37.9	0.250	0.241	2977.7	3.7	A	
76	2992.1	3029.8	37.6	0.250	0.246	3019.1	1.6	A	
77	3031.8	3069.9	38.2	0.250	0.241	3060.7	3.5	A	
78	3071.9	3111.4	39.5	0.250	0.244	3100.1	2.5	A	
79	3113.4	3151.9	38.4	0.250	0.243	3117.7	3.0	A	
80	3153.9	3191.8	37.9	0.250	0.244	3182.1	2.3	A	
81	3193.8	3232.2	38.4	0.250	0.243	3222.9	2.7	A	
82	3234.2	3273.2	39.0	0.250	0.245	3264.5	2.0	A	
83	3275.2	3314.2	39.0	0.250	0.245	3305.2	1.9	A	
84	3316.2	3355.2	39.0	0.250	0.244	3345.8	2.5	A	
85	3357.2	3395.6	38.4	0.250	0.244	3386.7	2.5	A	
86	3397.6	3436.1	38.4	0.250	0.241	3427.0	3.5	A	
87	3438.1	3475.1	37.1	0.250	0.242	3443.3	3.1	A	
88	3477.1	3516.7	39.5	0.250	0.242	3507.7	3.0	A	
89	3518.7	3556.6	37.9	0.250	0.243	3548.7	2.7	A	
90	3558.6	3597.8	39.3	0.250	0.243	3570.9	2.8	A	
91	3599.8	3638.0	38.2	0.250	0.238	3629.8	4.9	A	
92	3640.0	3679.2	39.3	0.250	0.240	3669.7	3.9	A	

#	Top Body (ft)	Bottom Body (ft)	Body Length (ft)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (ft)	Max Loss (%)	Grade	Damage Profile (% wall loss)
93	3681.2	3719.1	37.9	0.250	0.242	3710.6	3.2	A	
94	3721.1	3760.1	39.0	0.250	0.243	3751.9	2.9	A	
95	3762.1	3800.6	38.4	0.250	0.243	3792.7	2.6	A	
96	3802.6	3841.3	38.7	0.250	0.241	3833.8	3.6	A	
97	3843.3	3881.7	38.4	0.250	0.240	3874.6	3.8	A	
98	3883.7	3923.2	39.5	0.250	0.241	3916.1	3.6	A	
99	3925.2	3963.4	38.2	0.250	0.240	3956.6	4.0	A	
100	3965.4	4004.1	38.7	0.250	0.244	4002.7	2.4	A	
101	4006.1	4044.6	38.4	0.250	0.240	4038.2	3.9	A	
102	4046.6	4085.3	38.7	0.250	0.240	4078.9	4.0	A	
103	4087.3	4126.3	39.0	0.250	0.246	4120.4	1.4	A	
104	4128.3	4168.1	39.8	0.250	0.243	4161.1	2.8	A	
105	4170.1	4208.8	38.7	0.250	0.243	4170.9	2.6	A	
106	4210.8	4249.5	38.7	0.250	0.242	4243.6	3.4	A	
107	4251.5	4289.4	37.9	0.250	0.243	4284.6	3.0	A	
108	4291.4	4331.2	39.8	0.250	0.241	4293.5	3.5	A	
109	4333.2	4371.1	37.9	0.250	0.245	4366.1	1.9	A	
110	4373.1	4413.4	40.3	0.250	0.247	4406.7	1.3	A	
111	4415.4	4454.4	39.0	0.250	0.243	4447.6	3.0	A	
112	4456.4	4494.8	38.4	0.250	0.243	4488.9	3.0	A	
113	4496.8	4536.4	39.5	0.250	0.247	4529.1	1.2	A	
114	4538.4	4576.5	38.2	0.250	0.243	4539.3	2.8	A	
115	4578.5	4617.2	38.7	0.250	0.244	4611.7	2.3	A	
116	4619.2	4658.5	39.3	0.250	0.240	4620.4	4.0	A	
117	4660.5	4699.8	39.3	0.250	0.242	4692.9	3.4	A	
118	4701.8	4739.7	37.9	0.250	0.238	4733.6	4.9	A	
119	4741.7	4780.9	39.3	0.250	0.242	4775.0	3.2	A	
120	4782.9	4822.2	39.3	0.250	0.237	4815.8	5.3	B	
121	4824.2	4862.1	37.9	0.250	0.242	4856.5	3.1	A	
122	4864.1	4902.5	38.4	0.250	0.243	4897.2	2.7	A	
123	4904.5	4944.8	40.3	0.250	0.241	4906.5	3.4	A	
124	4946.8	4950.3	3.4	0.250	0.240	4947.7	4.1	A	
125	4952.3	4958.1	5.9	0.250	0.261	4956.7	-4.6	E	
126	4960.1	4964.1	4.0	0.250	0.243	4962.5	2.9	A	
127	4966.1	4970.4	4.2	0.250	0.243	4968.6	2.9	A	
128	4972.4	5010.5	38.2	0.250	0.239	4977.5	4.3	A	
129	5012.5	5050.7	38.2	0.250	0.238	5018.3	4.7	A	
130	5052.7	5091.7	39.0	0.250	0.239	5059.0	4.5	A	
131	5093.7	5133.2	39.5	0.250	0.236	5100.6	5.8	B	
132	5135.2	5173.6	38.4	0.250	0.242	5141.9	3.4	A	
133	5175.6	5179.6	4.0	0.250	PUP				
134	5181.6	5187.2	5.6	0.250	PDMS				
135	5189.2	5200.8	11.6	0.250	PUP-NIPPLE				
136	5202.8	5206.5	3.7	0.250	PUP				
137	5208.5	5246.9	38.4	0.250	0.248	5224.2	0.7	A	

#	Top Body (ft)	Bottom Body (ft)	Body Length (ft)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (ft)	Max Loss (%)	Grade	Damage Profile (% wall loss)
138	5248.9	5252.6	3.7	0.250	PUP				
139	5254.6	5272.7	18.1	0.250	PBR Assembly				
140	5274.7	5278.9	4.2	0.250	PUP				
141	5280.9	5284.1	3.2	0.250	PUP				
142	5286.1	5298.2	12.1	0.250	PACKER				
143	5300.2	5303.1	2.9	0.250	PUP JOINT				
144	5305.1	5344.1	39.0	0.250	0.233	5308.6	7.0	B	
145	5346.1	5385.3	39.3	0.250	0.230	5346.3	7.8	B	
146	5387.3	5392.7	5.3	0.250	PUP-NIPPLE				
147	5394.7	5397.8	3.2	0.250	PUP				
148	5399.8	5405.0	5.2	0.250	HALF MULE SHOE GUIDE				
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#	Top Body (ft)	Bottom Body (ft)	Body Length (ft)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (ft)	Max Loss (%)	Grade	Damage Profile (% wall loss)
183									
184									
185									
186									
187									
188									
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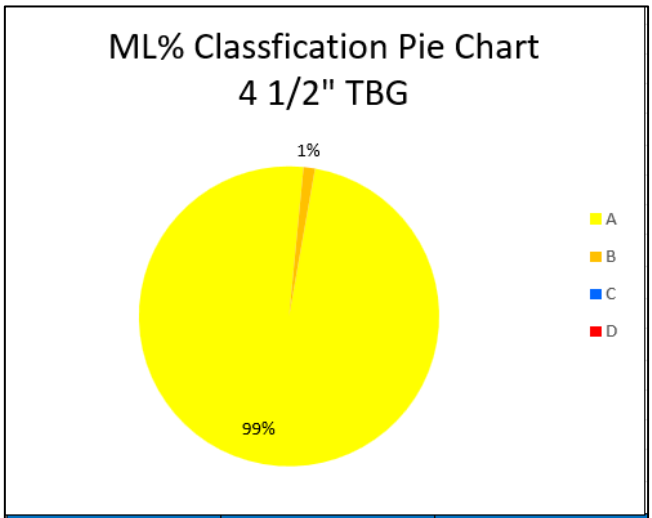
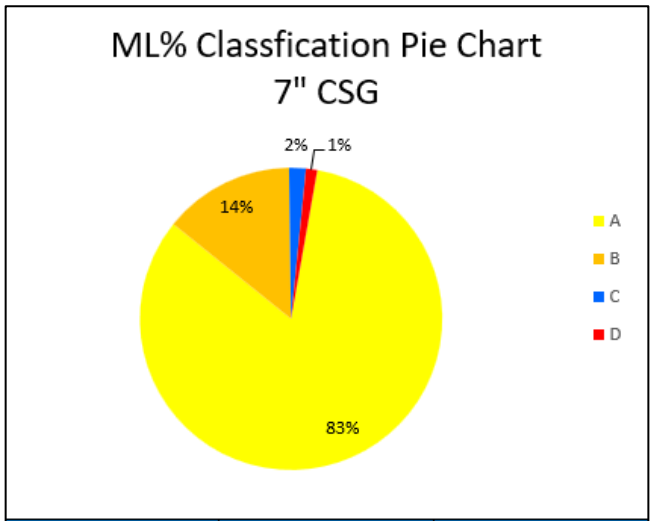
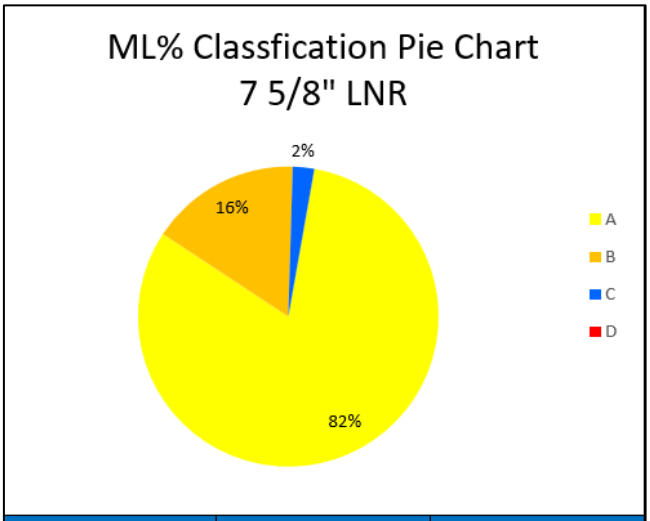
13. Metal Loss Pie Chart Summary



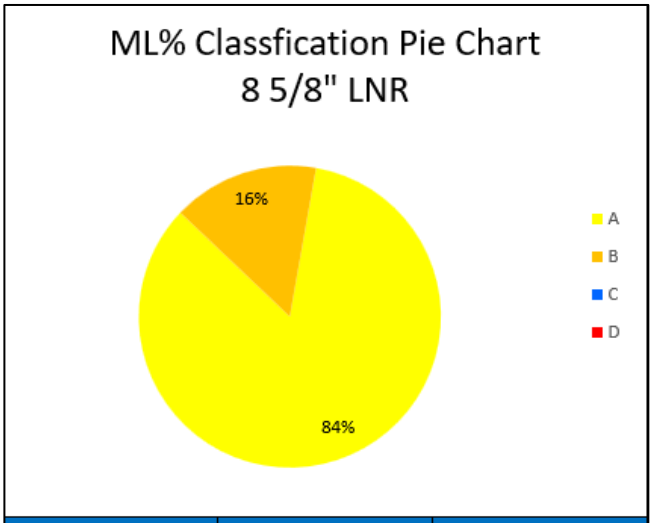
Grade	Joints	% of Total
A	159	99%
B	2	1%
C	0	0%
D	0	0%
Total	161	100%



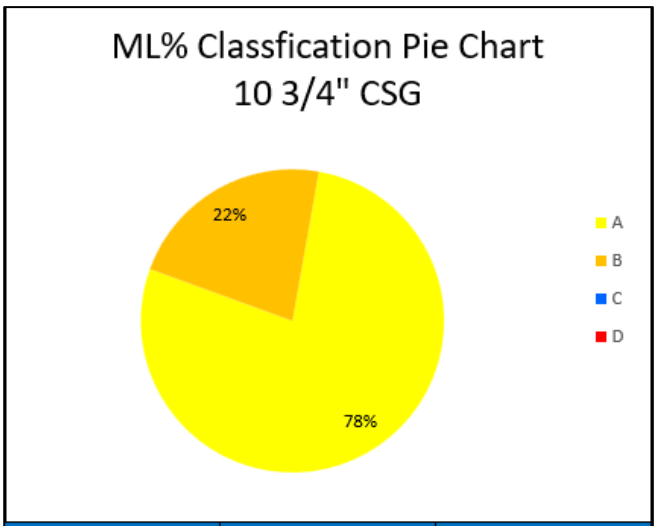
Grade	Joints	% of Total
A	139	83%
B	23	14%
C	3	2%
D	2	1%
Total	167	100%



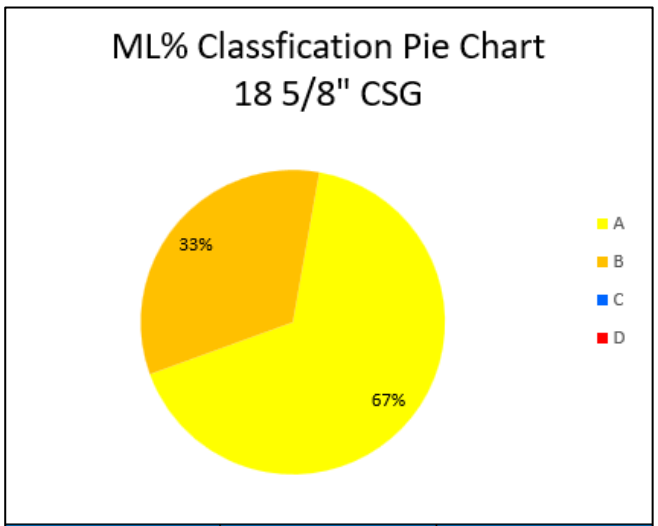
Grade	Joints	% of Total
A	35	82%
B	7	16%
C	1	2%
D	0	0%
Total	43	100%



Grade	Joints	% of Total
A	43	84%
B	8	16%
C	0	0%
D	0	0%
Total	51	100%



Grade	Joints	% of Total
A	7	78%
B	2	22%
C	0	0%
D	0	0%
Total	9	100%



Grade	Joints	% of Total
A	2	67%
B	1	33%
C	0	0%
D	0	0%
Total	3	100%

Figure 14 - Metal Loss Pie Chart Summary

14. Quality Control

Data quality control was applied as per standard operating procedure and Saudi Aramco guidelines, for raw data for both main and repeat passes.

Quality control curves for the IL, IM and IS (long, medium, and short spacing coils currents) are within 700 mA and 500 mA.

- S, M, L polarities alternate from 0 - 1.
- Equivalent ADEC curves, with 60 channels, are derived from combining the curves of the three long, medium, and short coils.

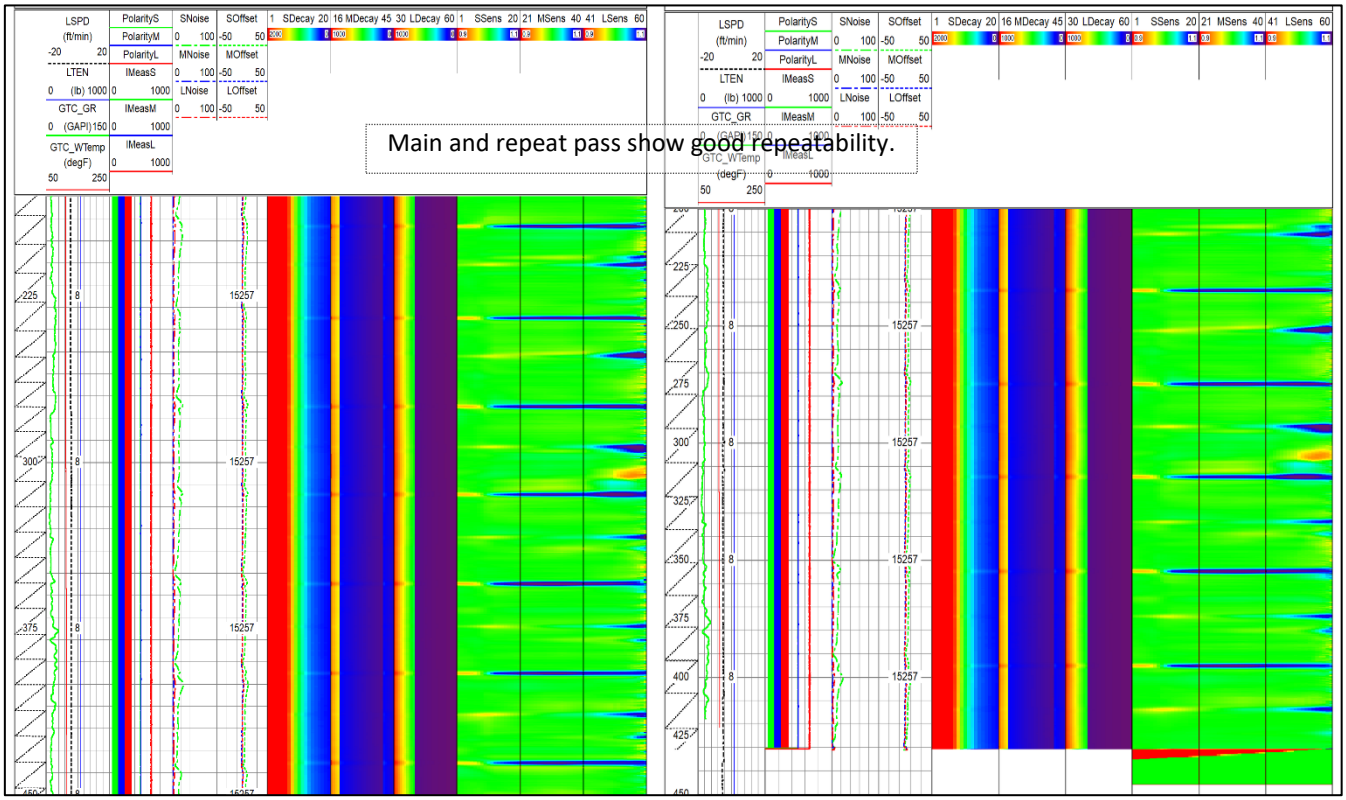


Figure 15 - Raw Data Repeatability

Quality control guidelines across the main pass is shown in the figure below.

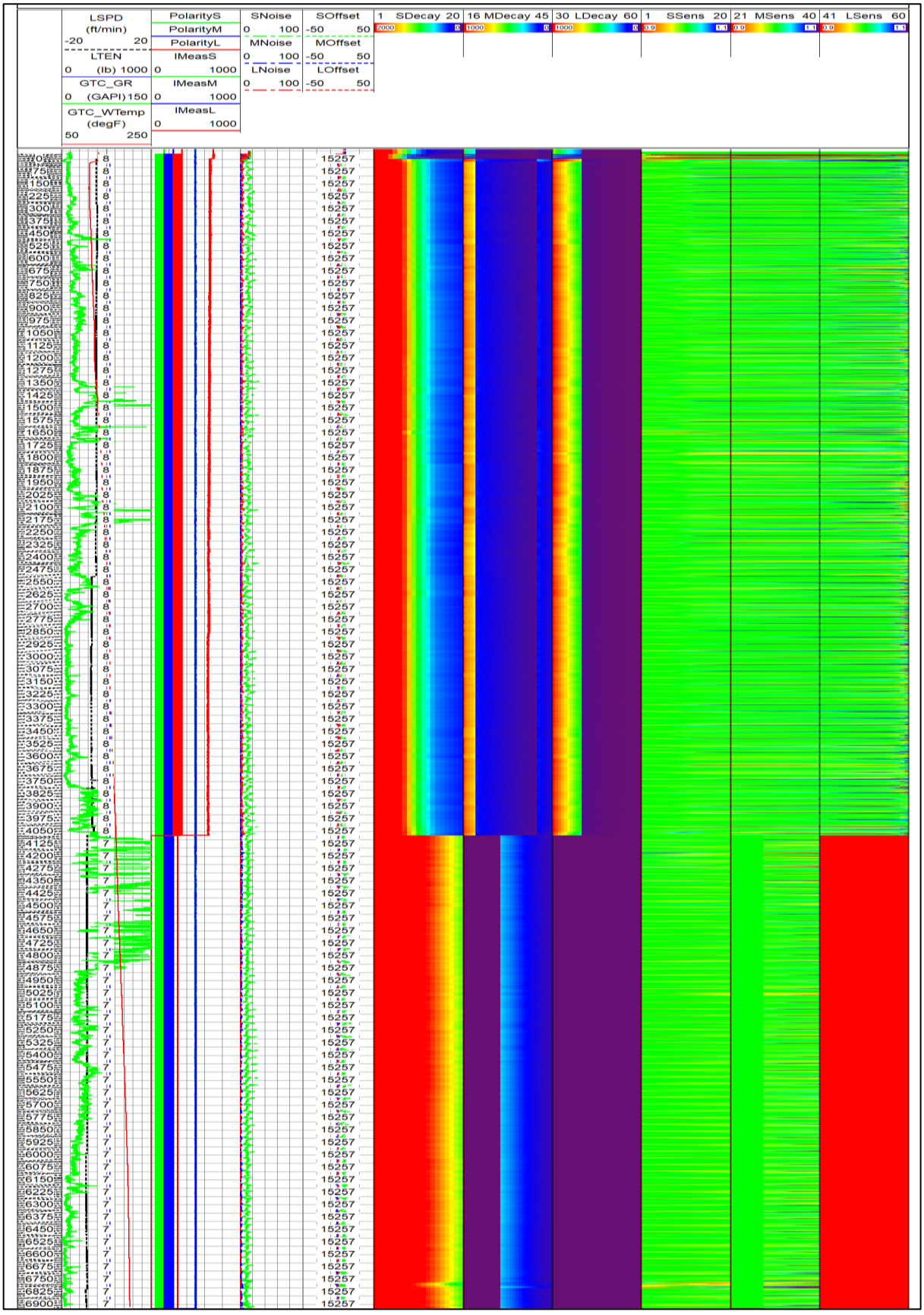


Figure 16 - Main Pass Quality Control

15. Well Completion Diagram

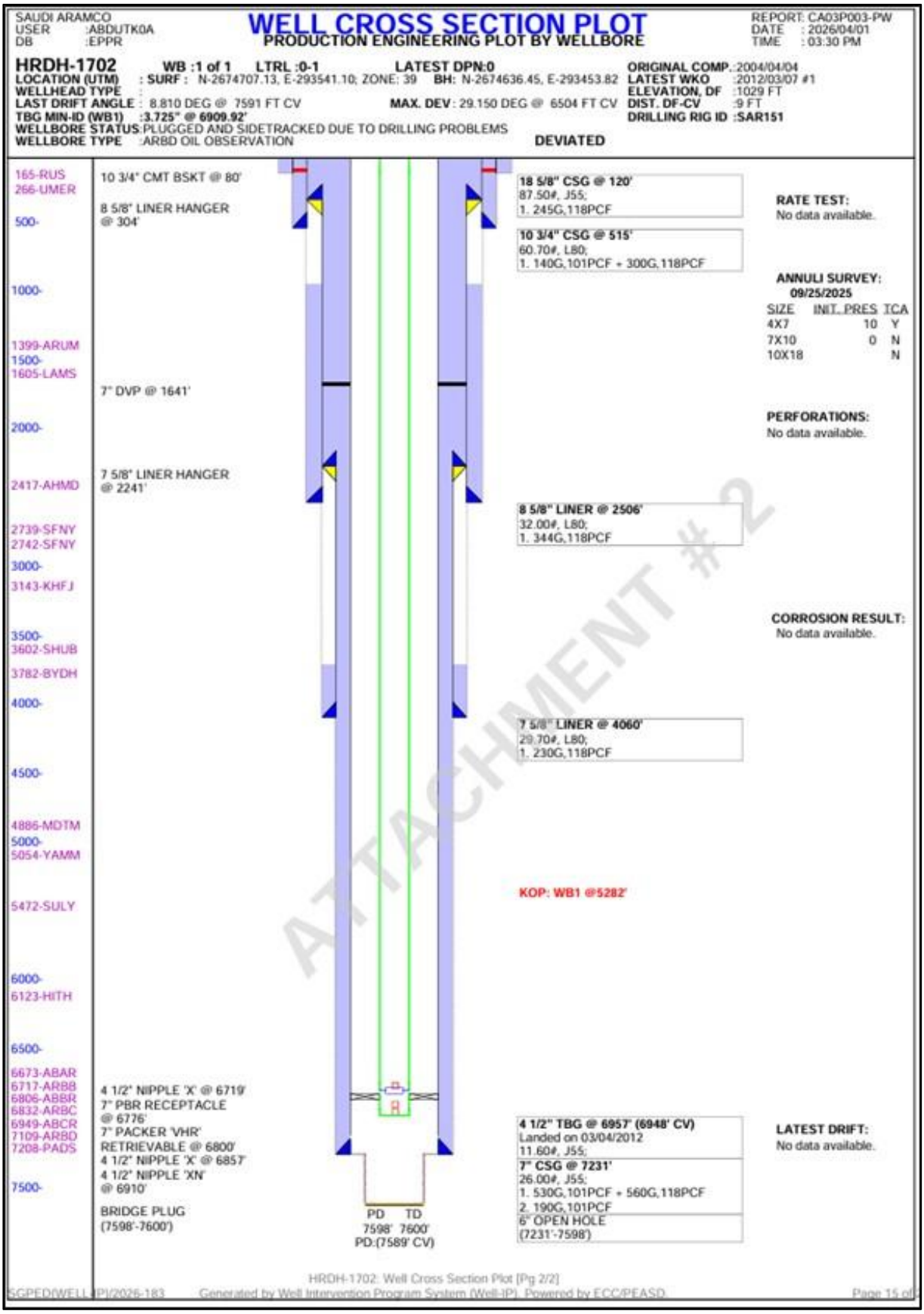


Figure 17 - Well Cross Section Plot

16. Tool String Diagram

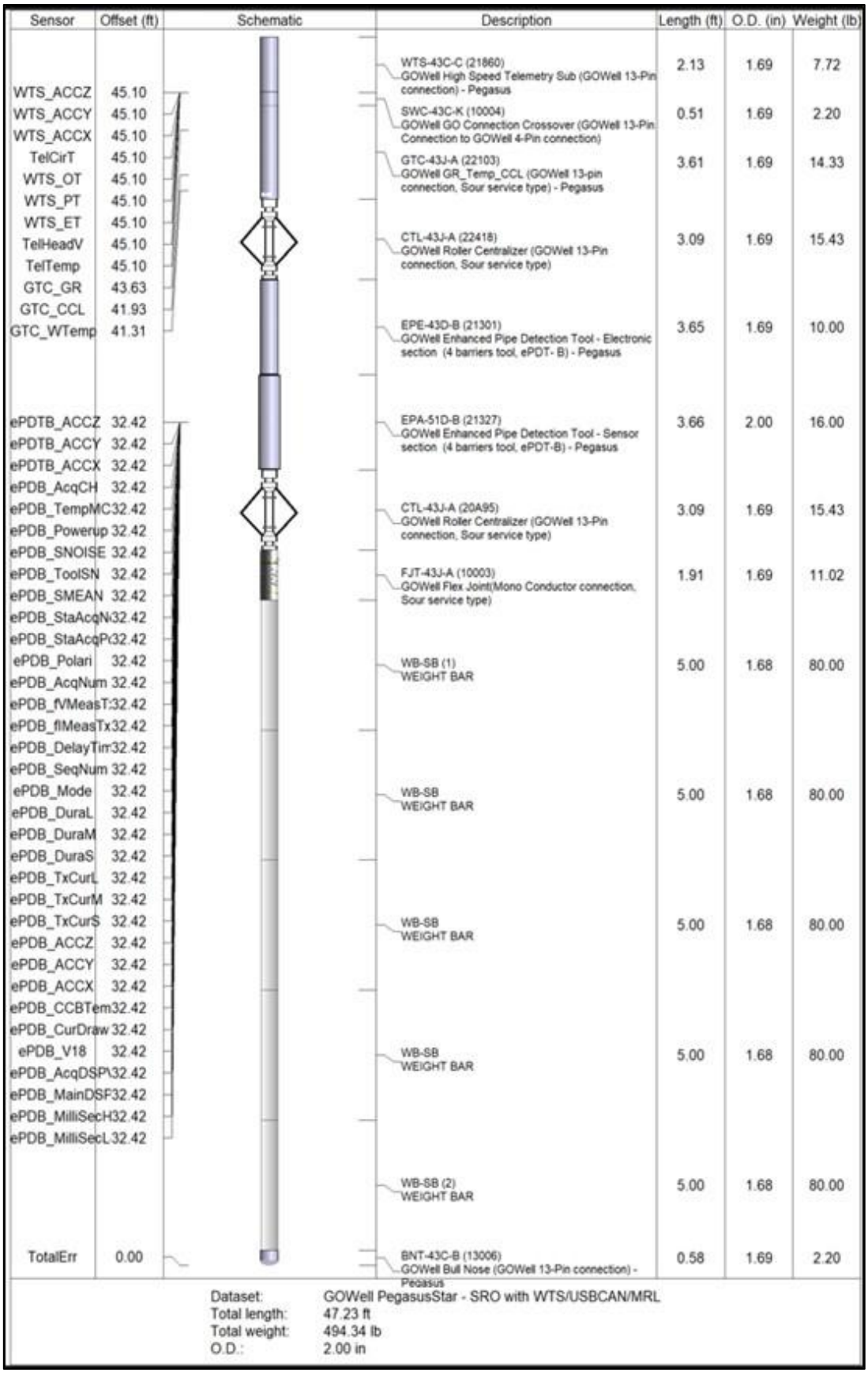


Figure 18 – Tool String Diagram

17. Appendix – Findings and Observations

- This appendix summarizes the key irregularities, data-quality issues, and operational challenges encountered during the acquisition, processing, and interpretation of the corrosion logging dataset. The objective is to provide transparency on factors that may influence data reliability, measurement accuracy, and the overall confidence level of the evaluation.

1. A. Processing Observations

1. Missing Data

- Tool stuck at 6990 ft (DF) and MSP was applied at the end of tubing after PE approval. We went up and down multiple times on OH sections with different speeds (60, 80, 100, 140, 160 ft/min) which resulted in missing recording the data from 6975 to 6925 ft.

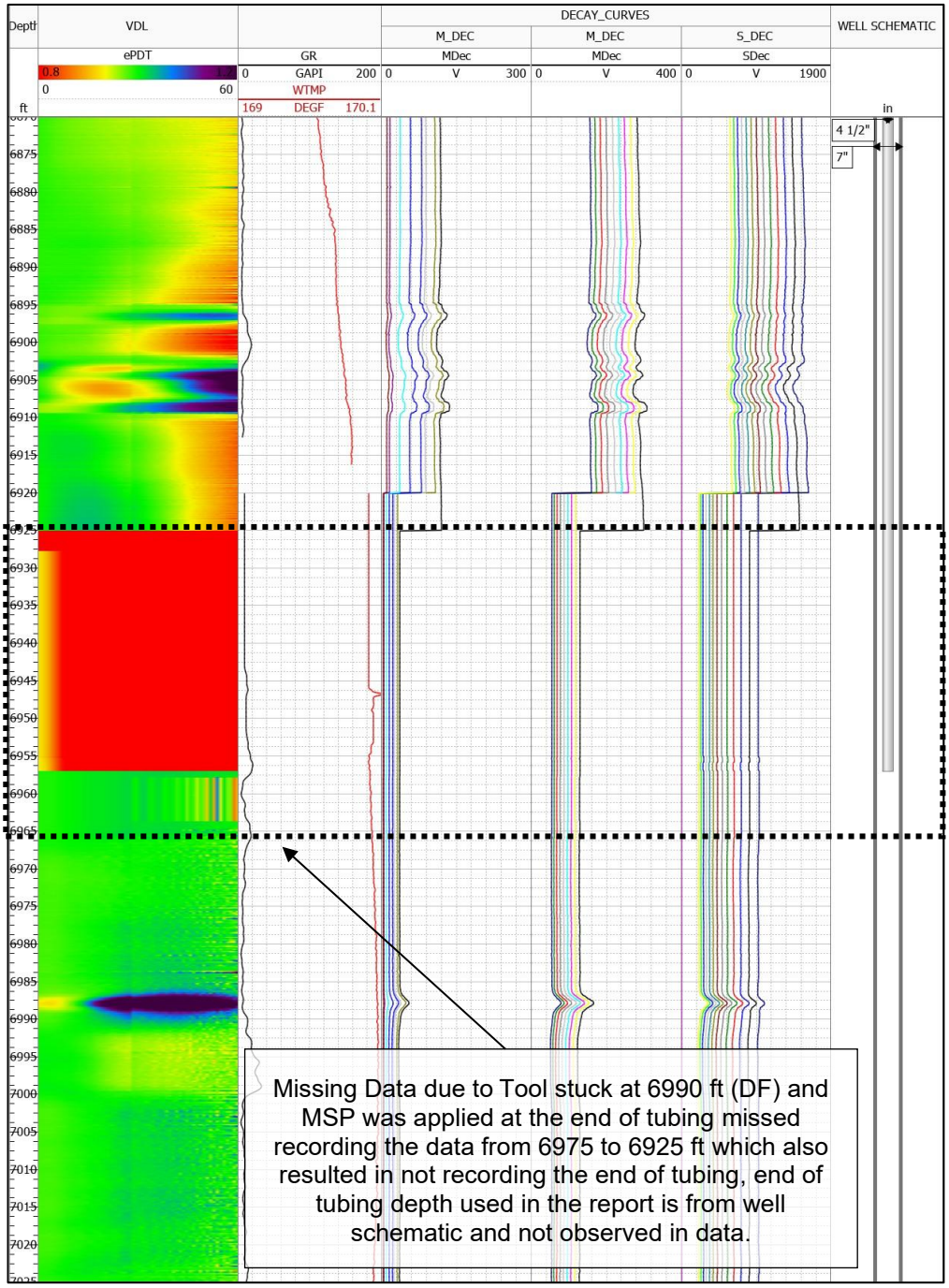


Figure 19 – Processing Observations